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(54) **METAL-INSULATOR-METAL CAPACITOR AND METHODS OF MANUFACTURING**

(71) Applicant: **Taiwan Semiconductor Manufacturing Company, Ltd.**, Hsinchu (TW)

(72) Inventors: **Yuan-Sheng HUANG**, Taichung City (TW); **Kaochao CHEN**, Hsinchu City (TW)

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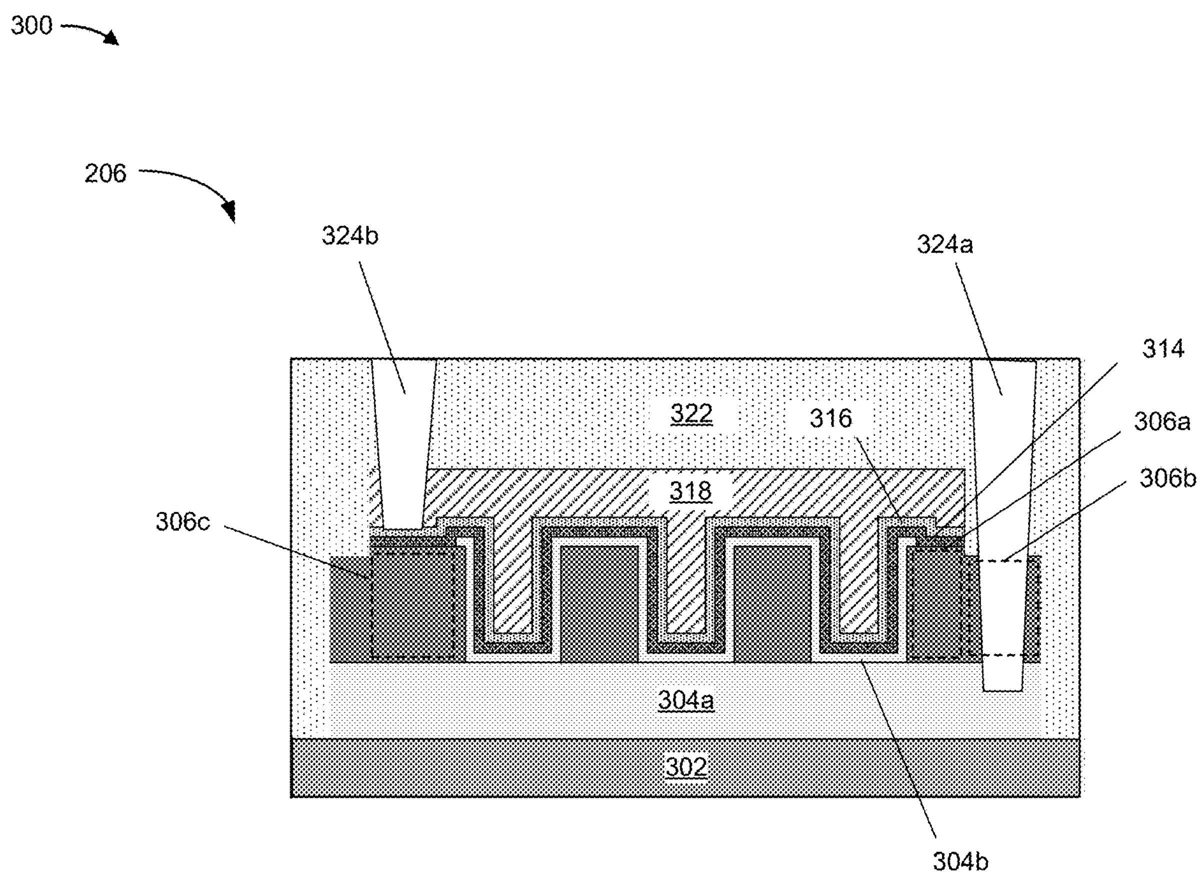
CPC ***H01L 23/528*** (2013.01)

(57) **ABSTRACT**

Some implementations described herein provide techniques and apparatuses for forming a semiconductor device including a metal-insulator-metal capacitor. The metal-insulator-metal capacitor includes a dielectric pad layer having a portion between a capacitor bottom metal electrode layer and a portion of an insulator layer. The dielectric pad layer may preserve a thickness of the insulator layer to reduce a likelihood of a leakage between a capacitor top metal electrode layer and the capacitor bottom metal electrode layer. The dielectric pad layer may also enable a reduction in a thickness of the insulator layer to increase a capacitance of the metal-insulator-metal capacitor.



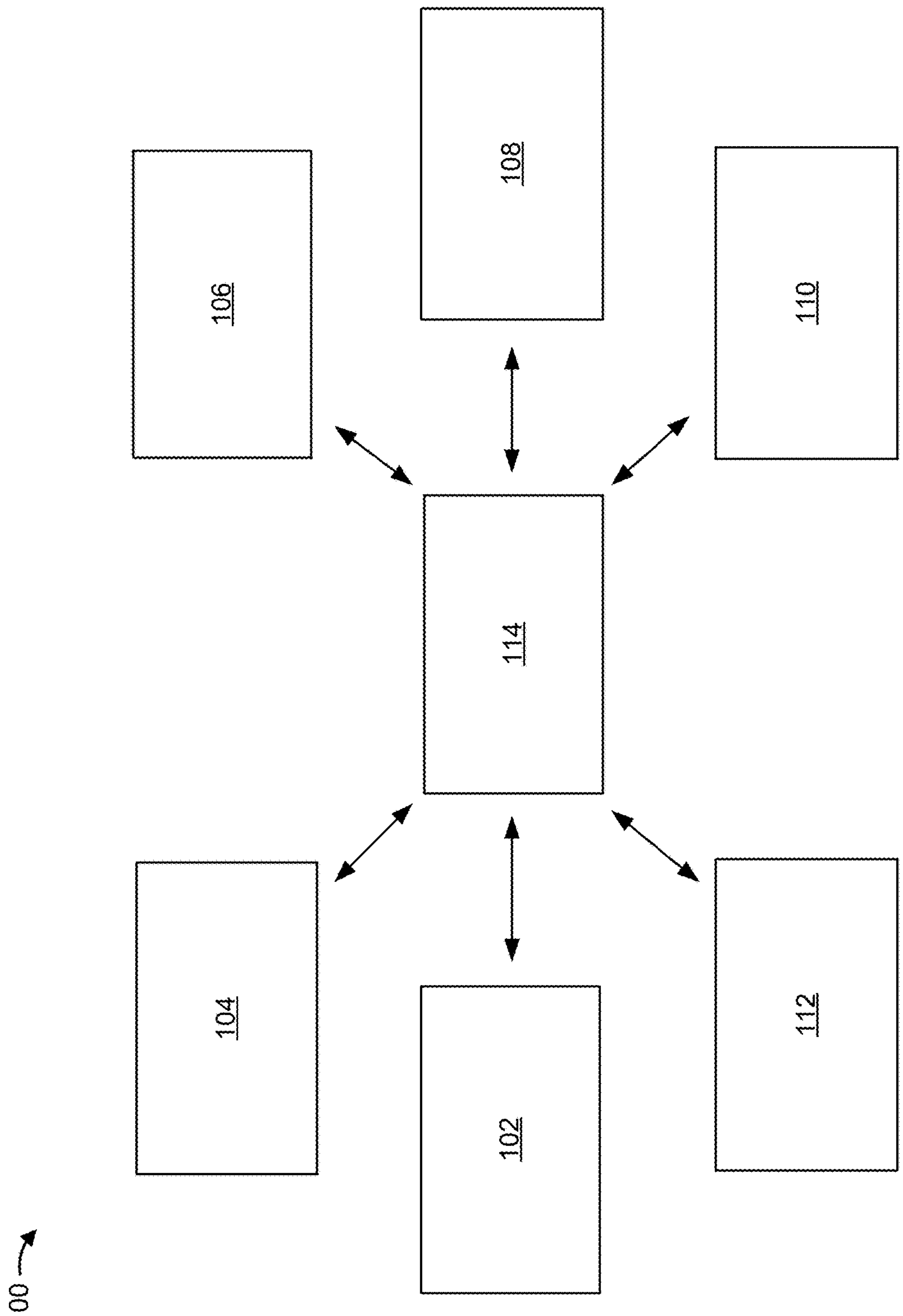


FIG. 1

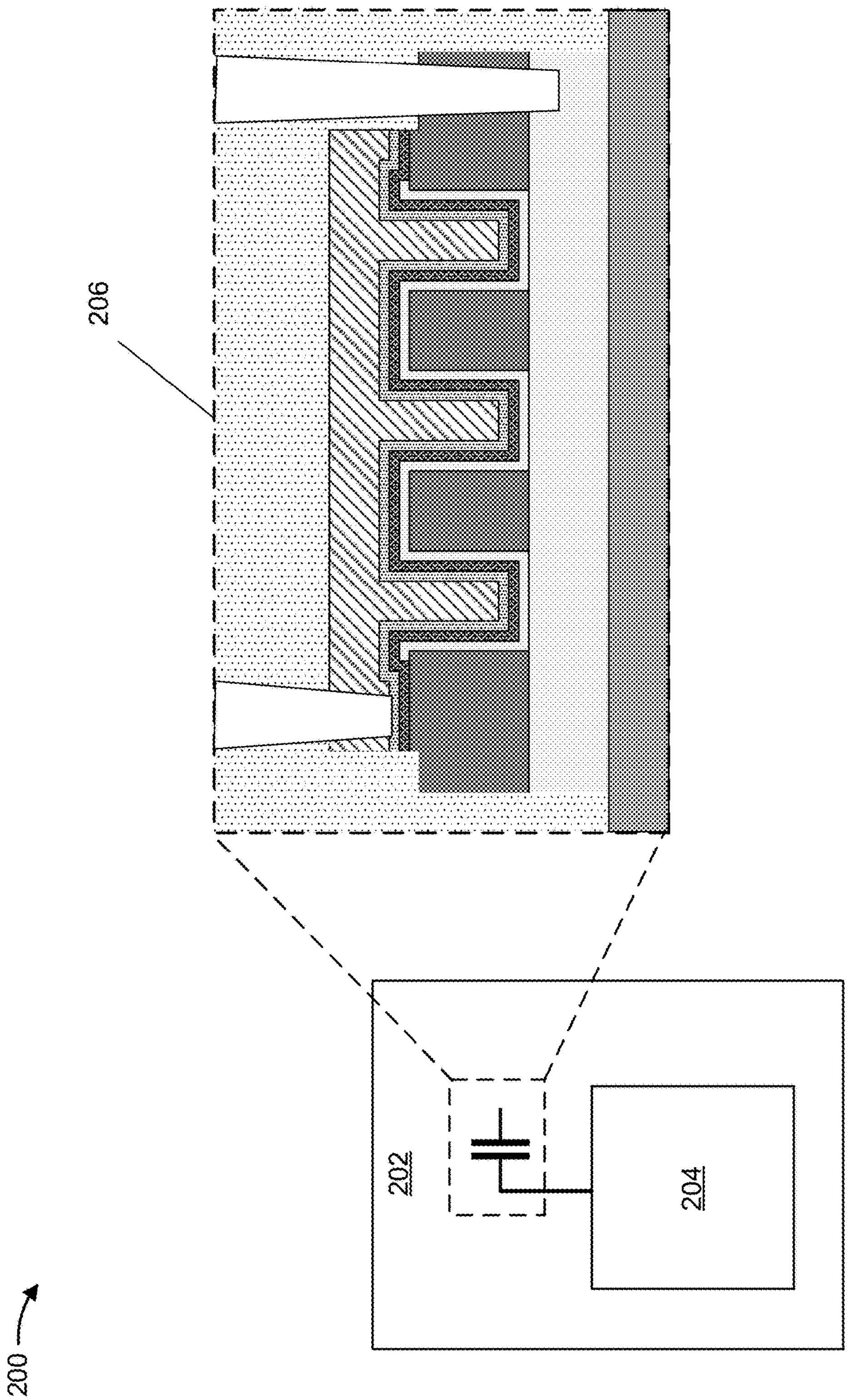


FIG. 2

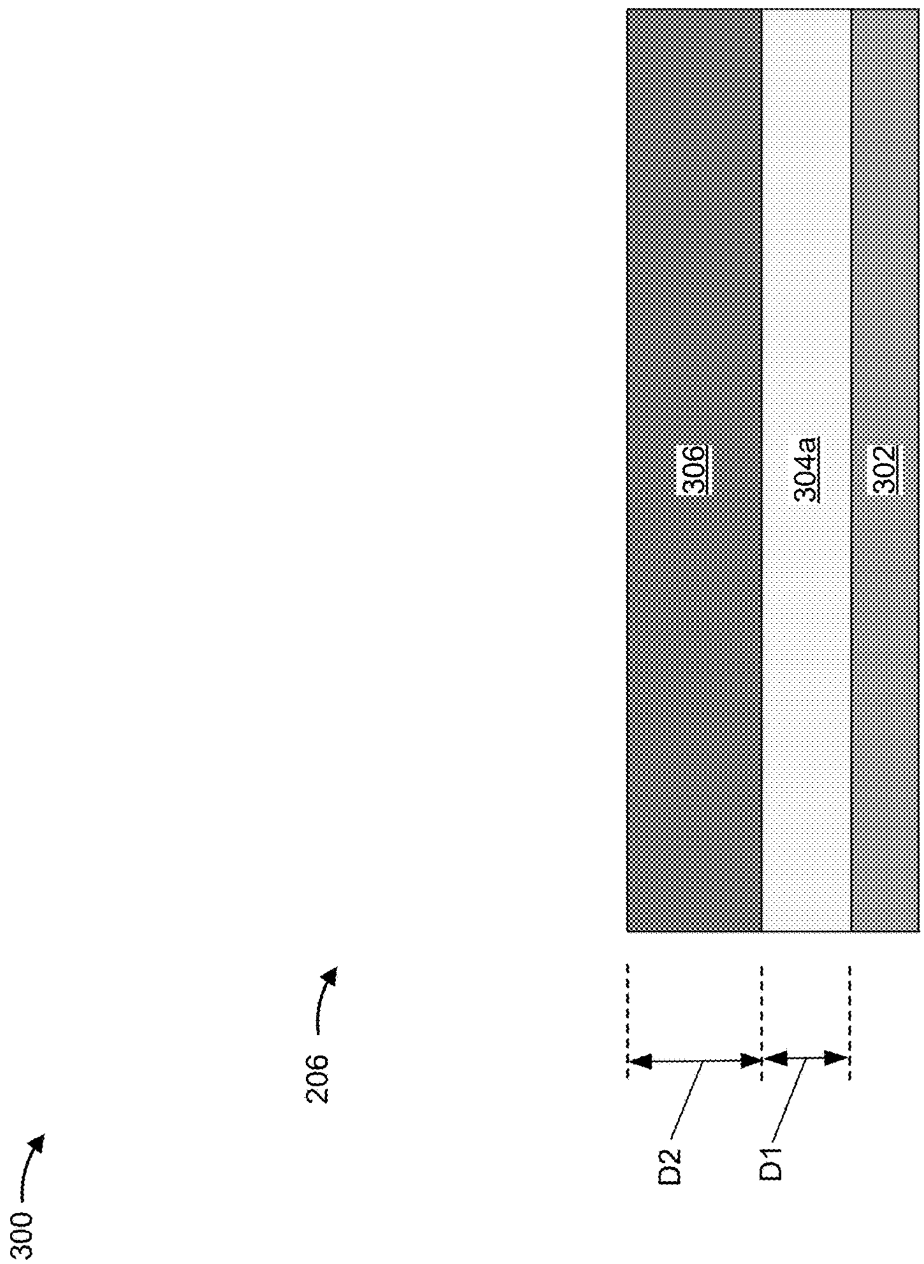


FIG. 3A

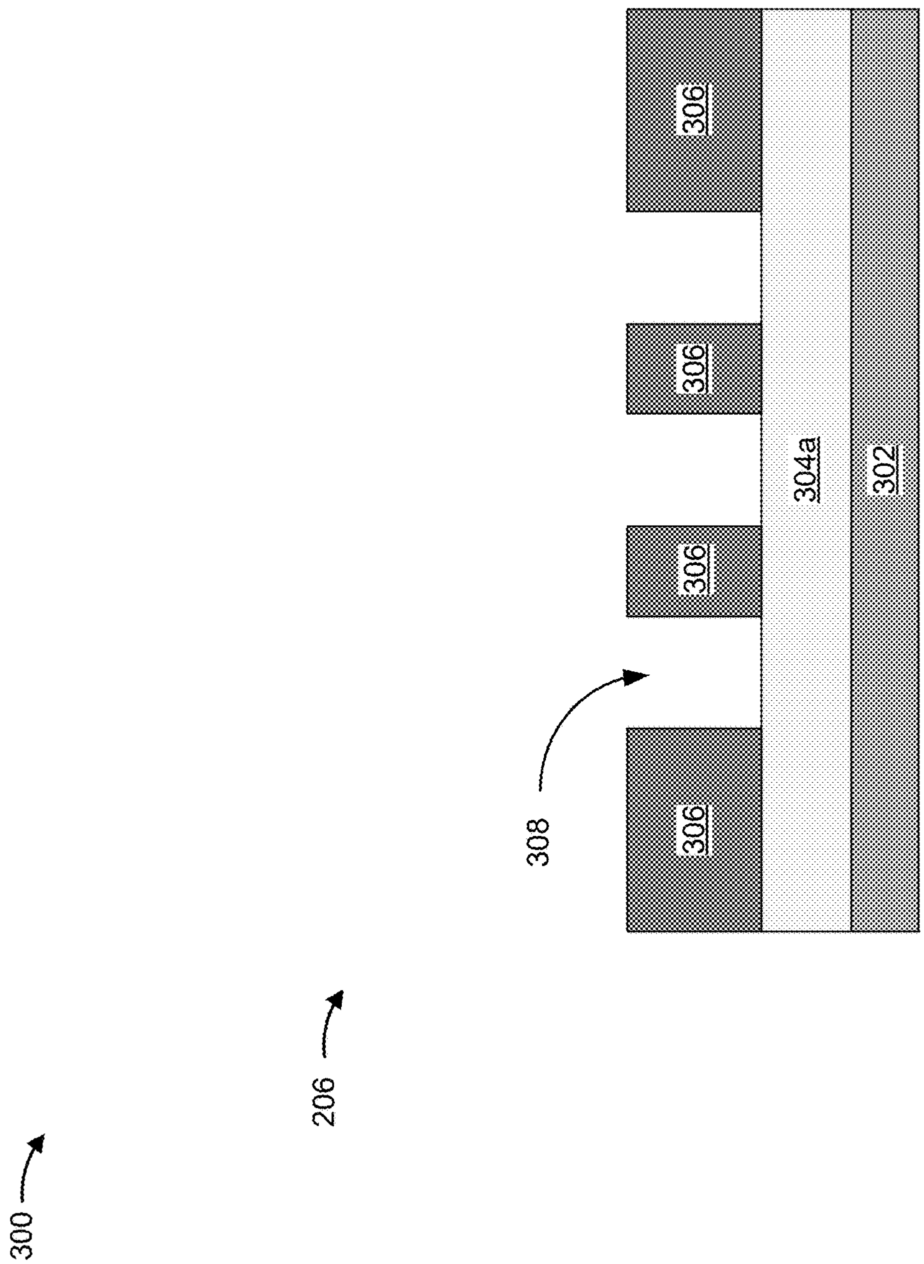


FIG. 3B

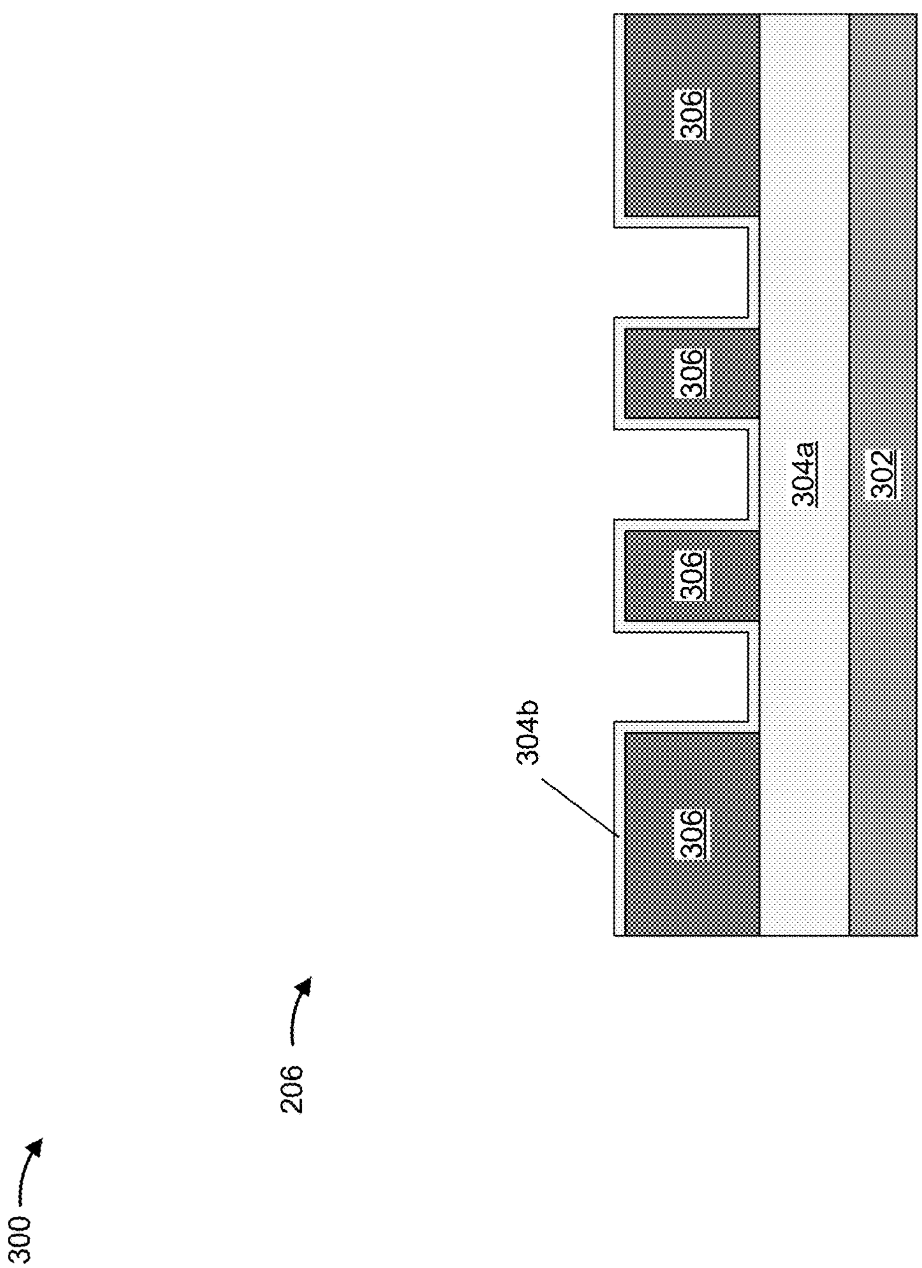


FIG. 3C

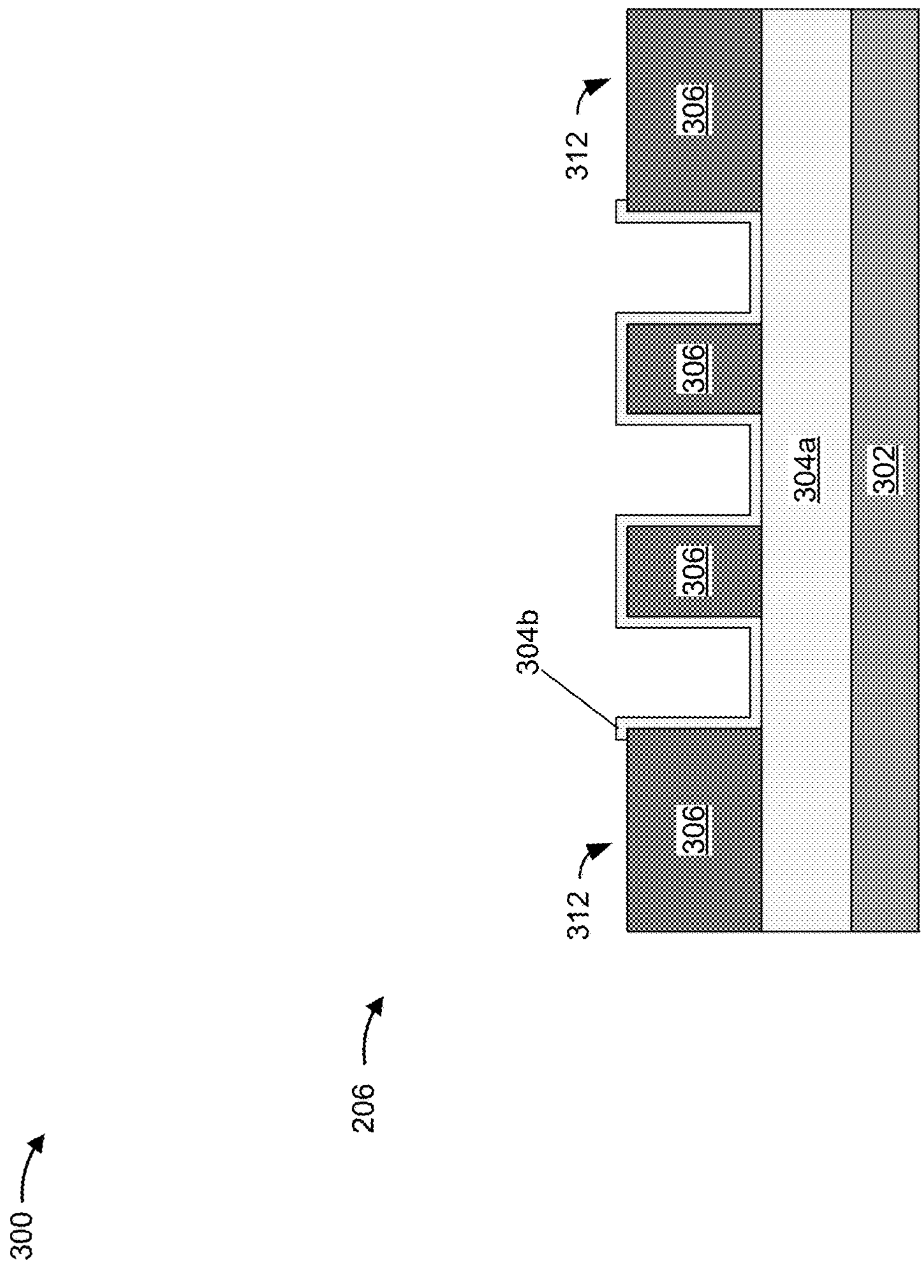


FIG. 3D

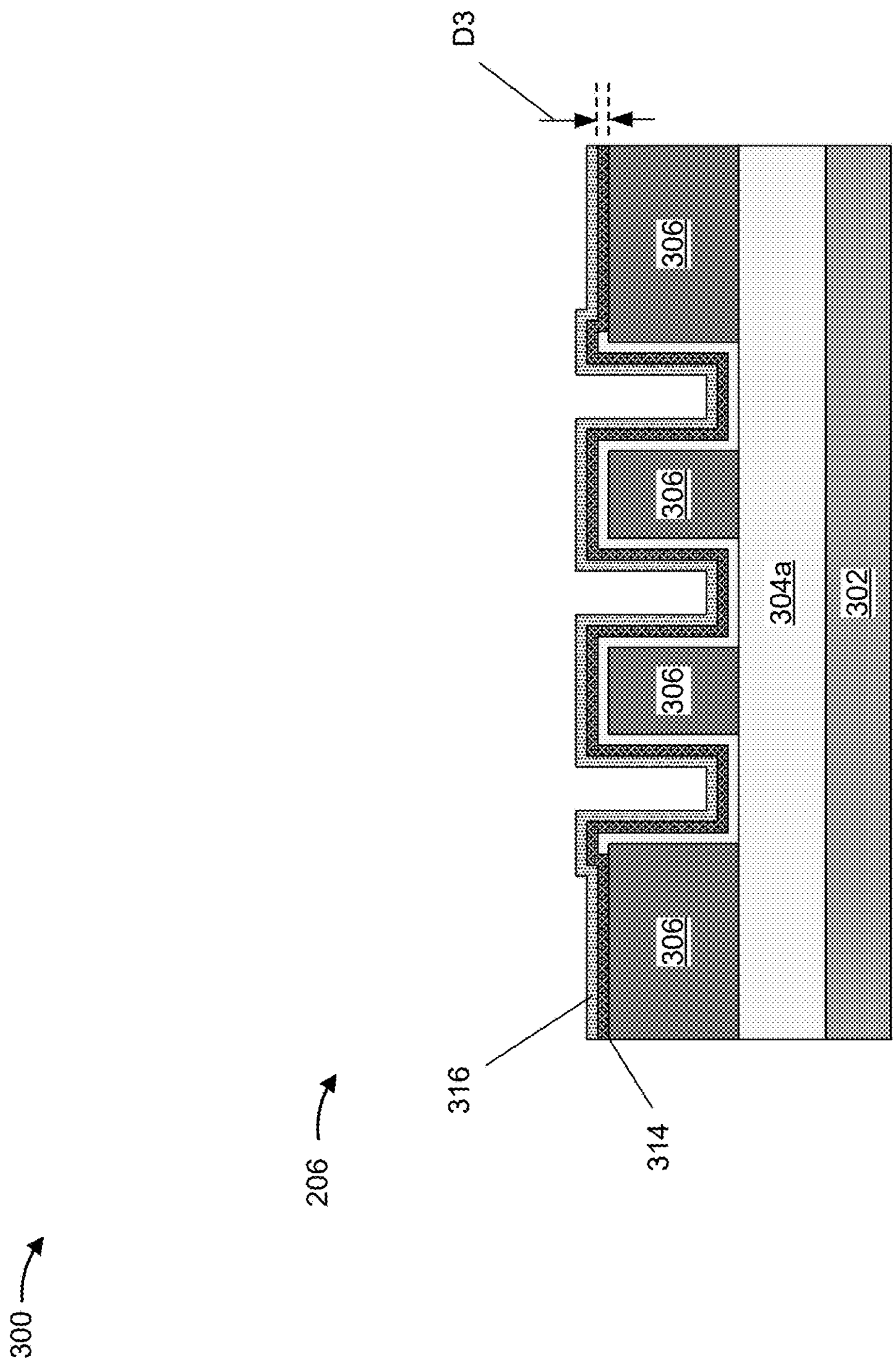


FIG. 3E

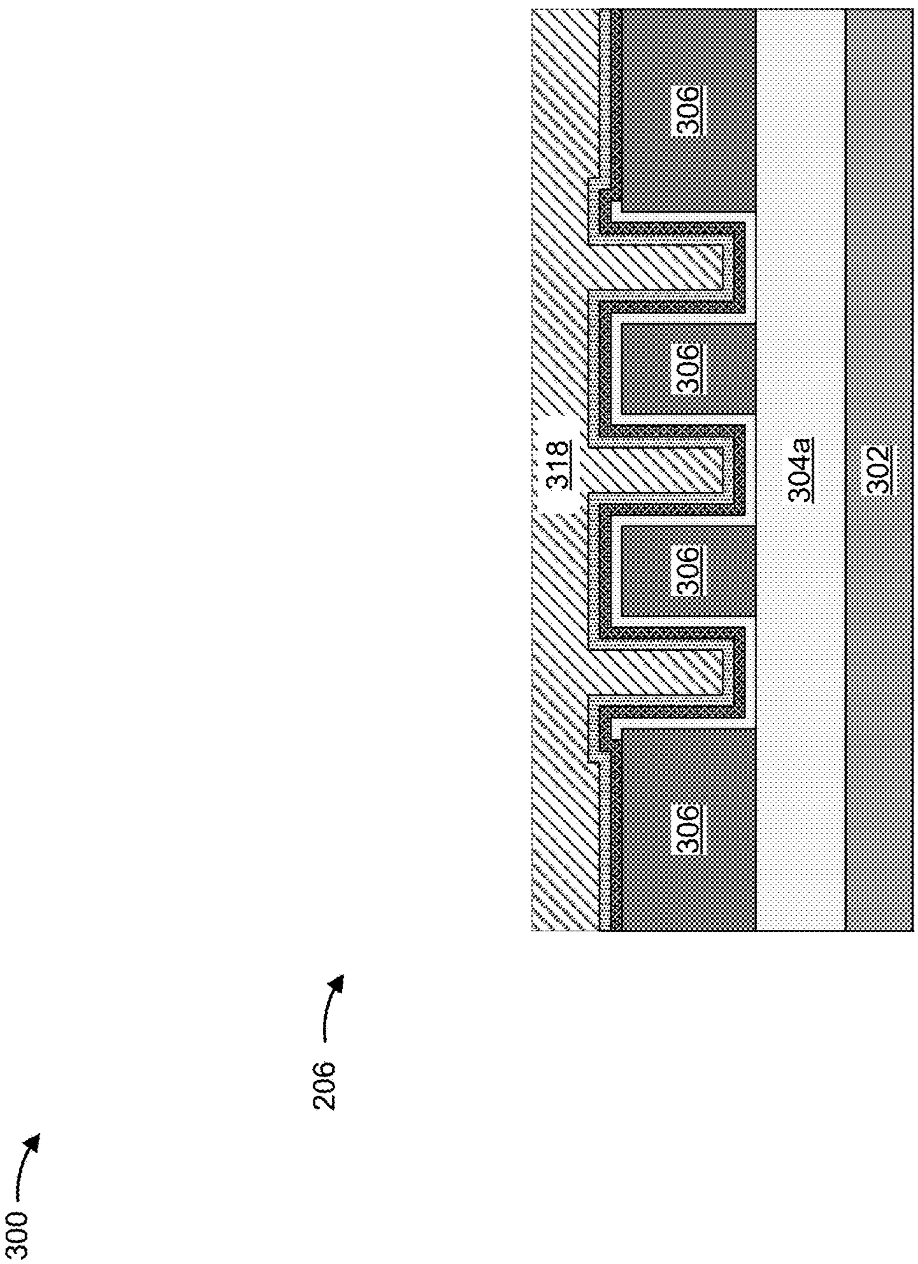


FIG. 3F

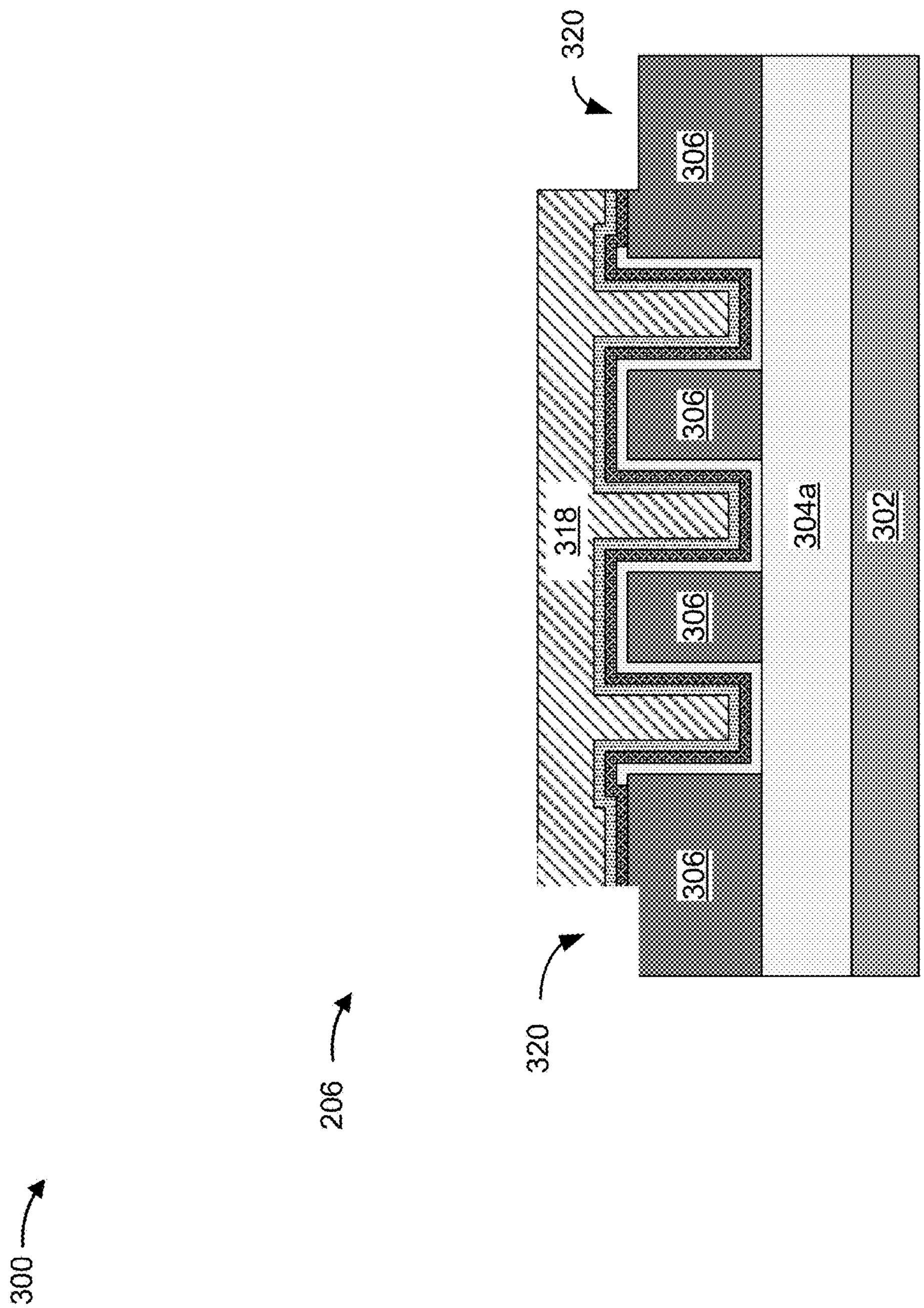


FIG. 3G

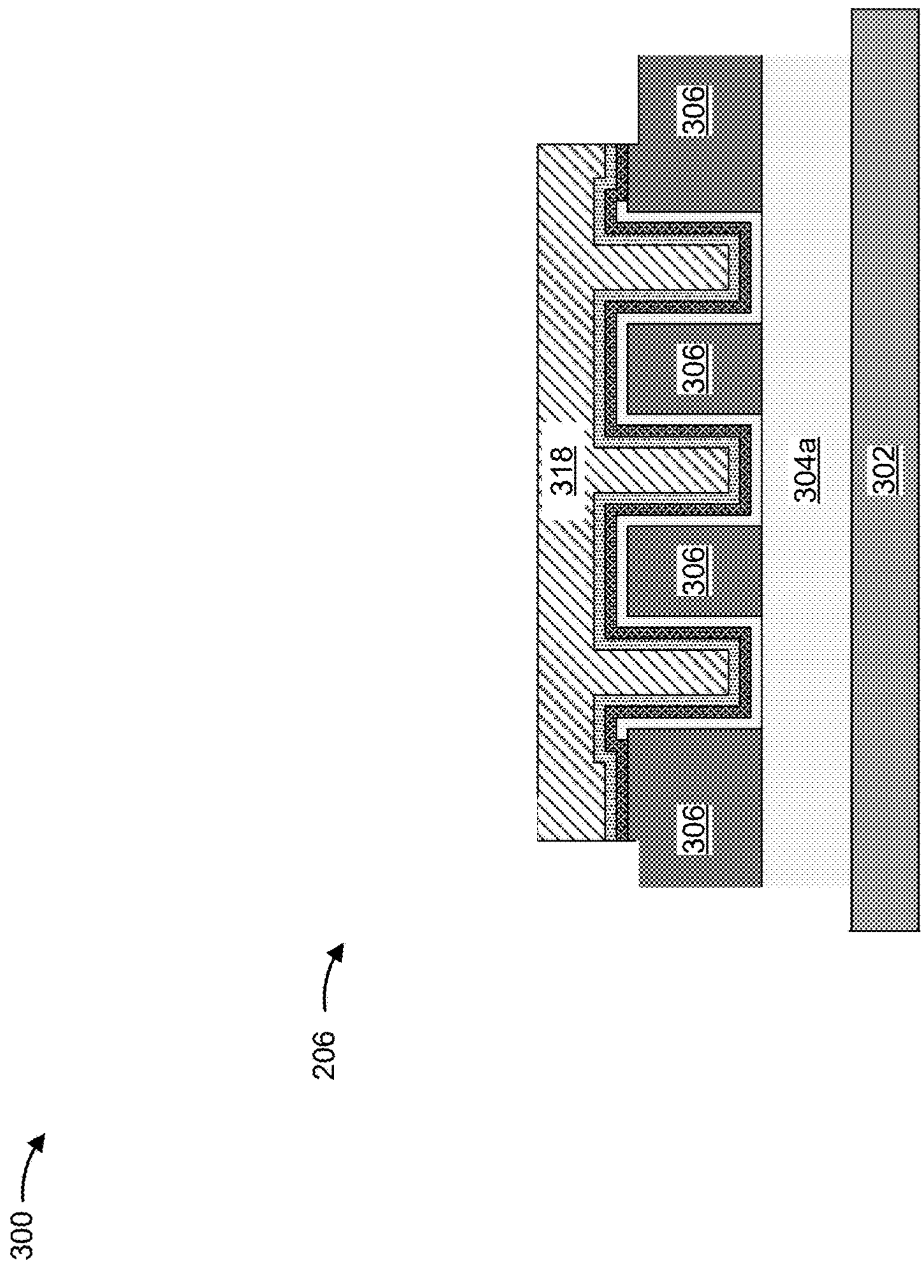


FIG. 3H

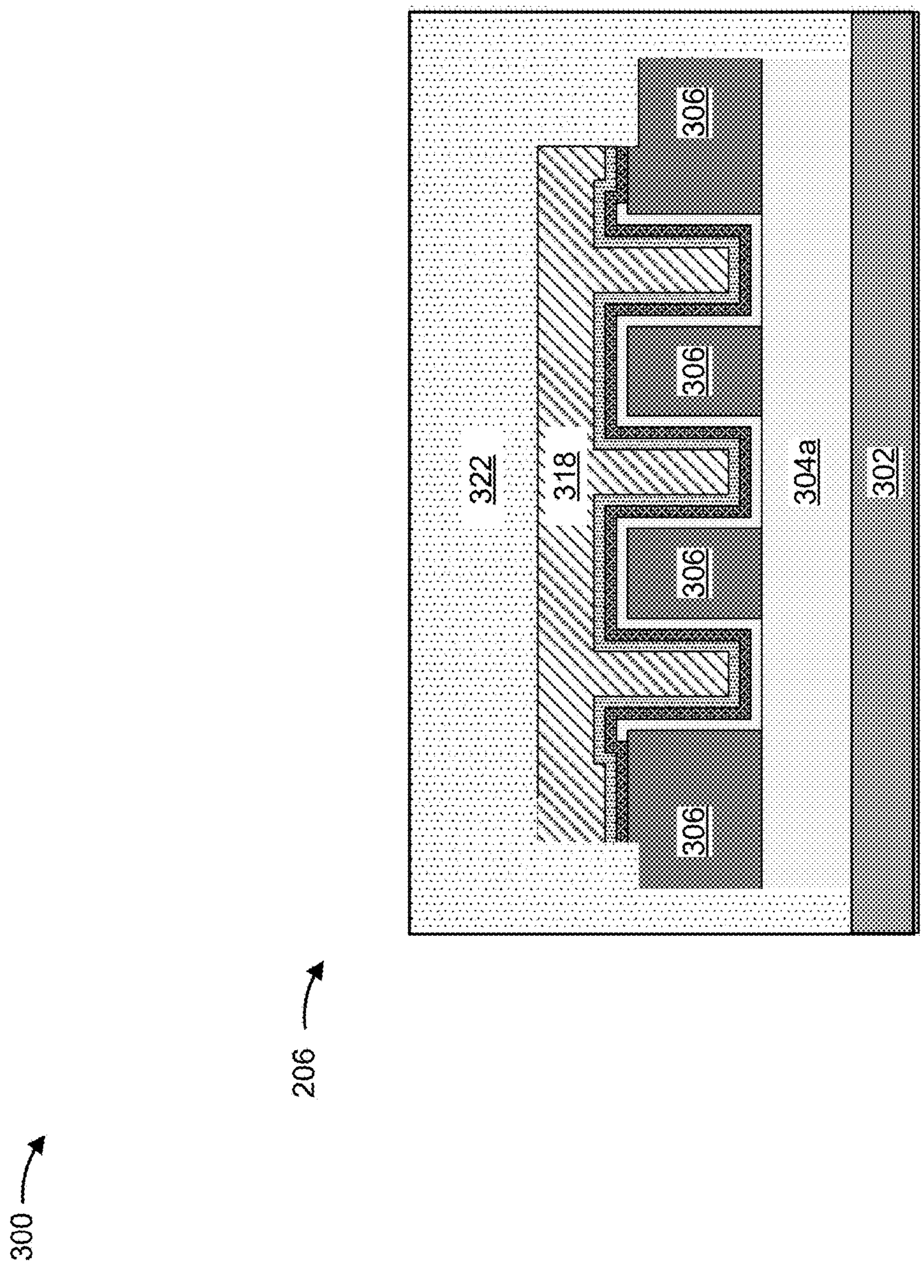


FIG. 3I

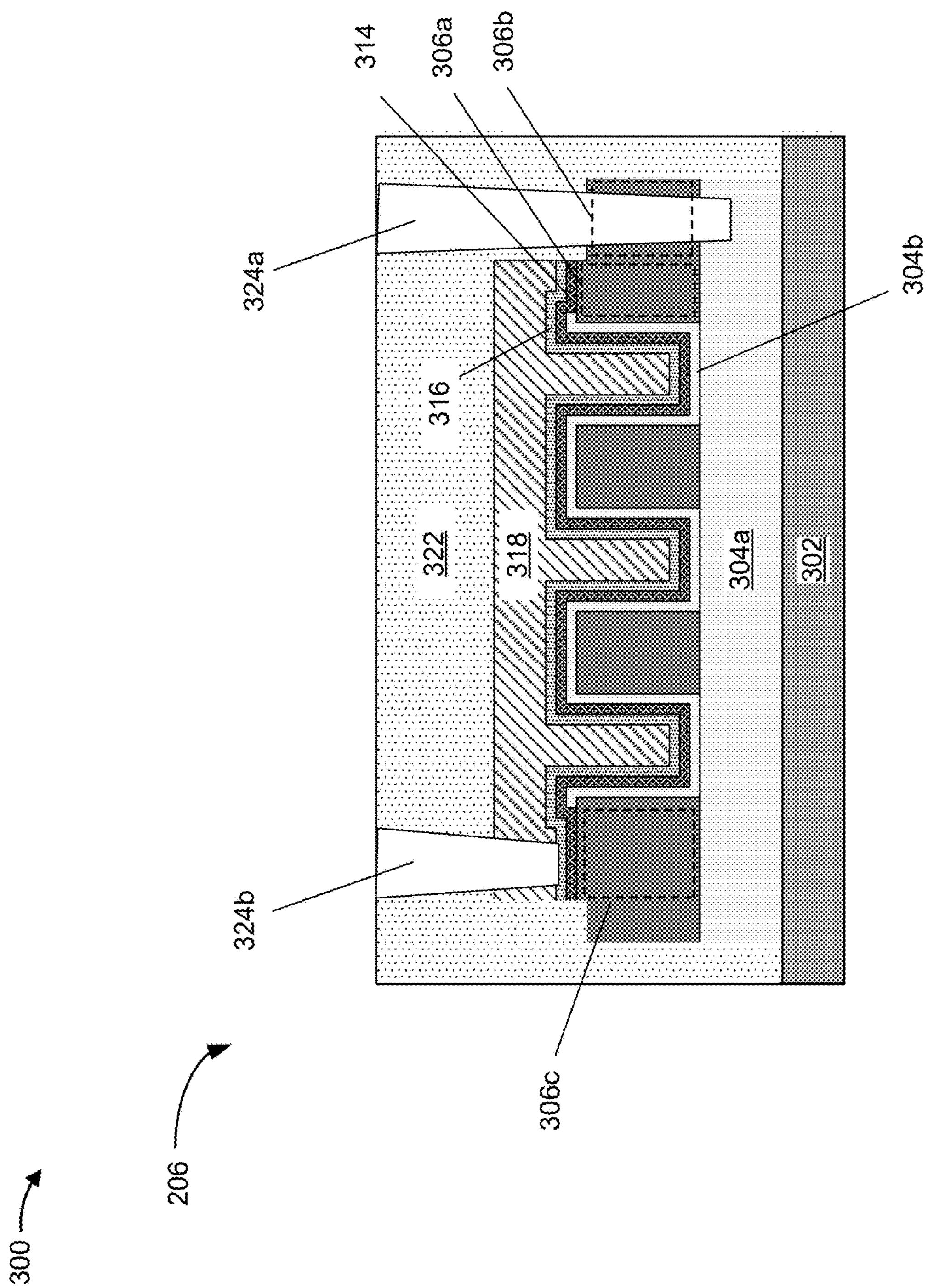


FIG. 3J

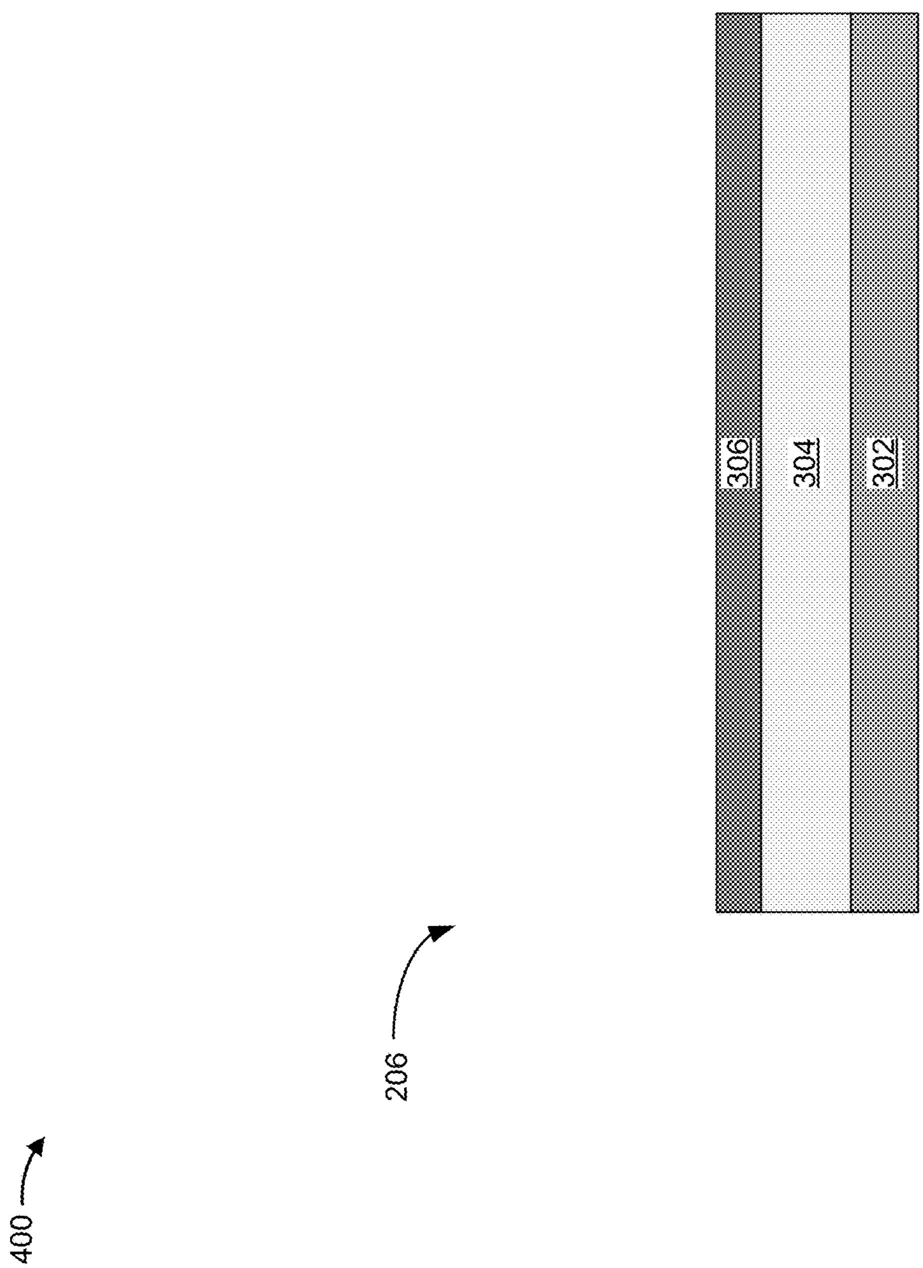


FIG. 4A

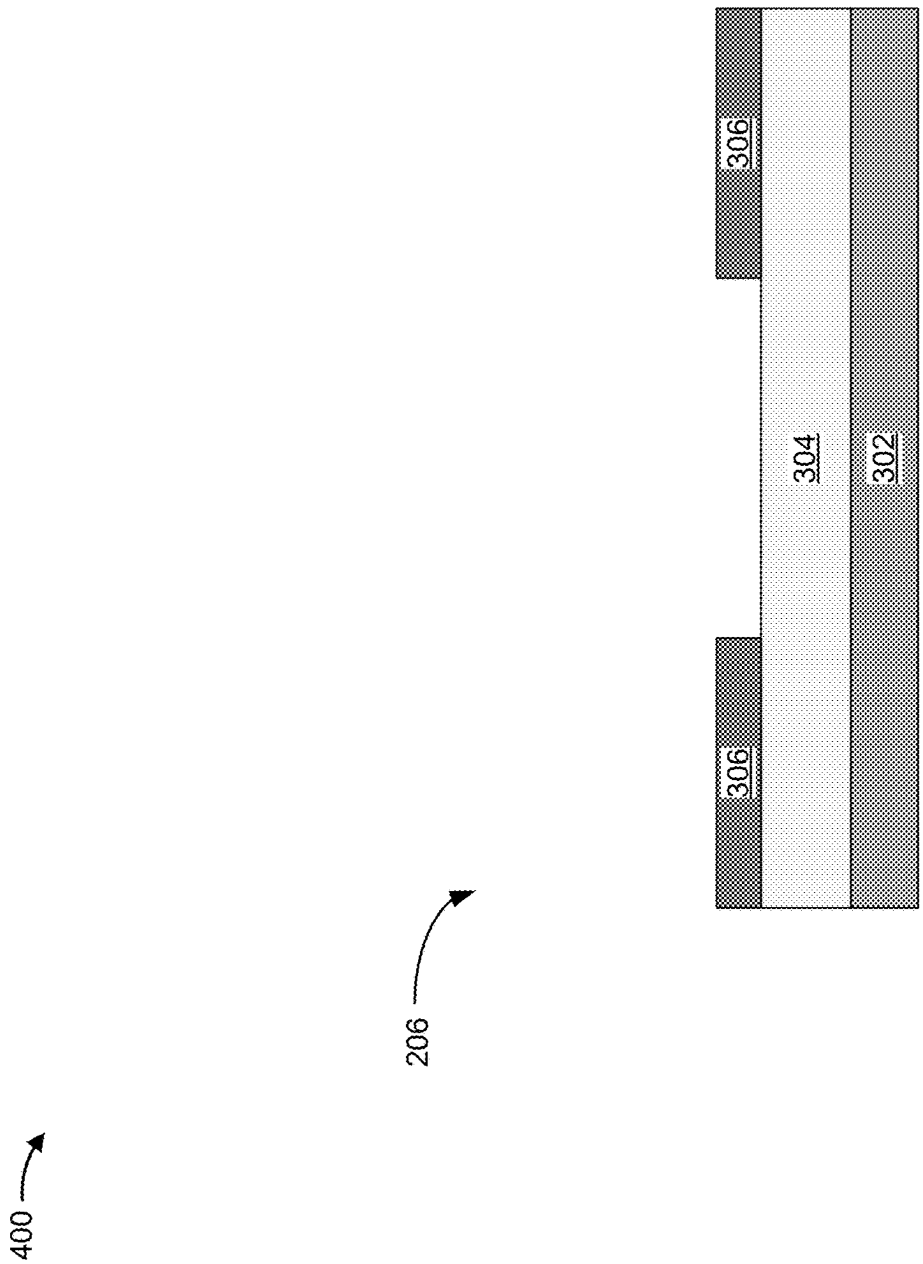


FIG. 4B

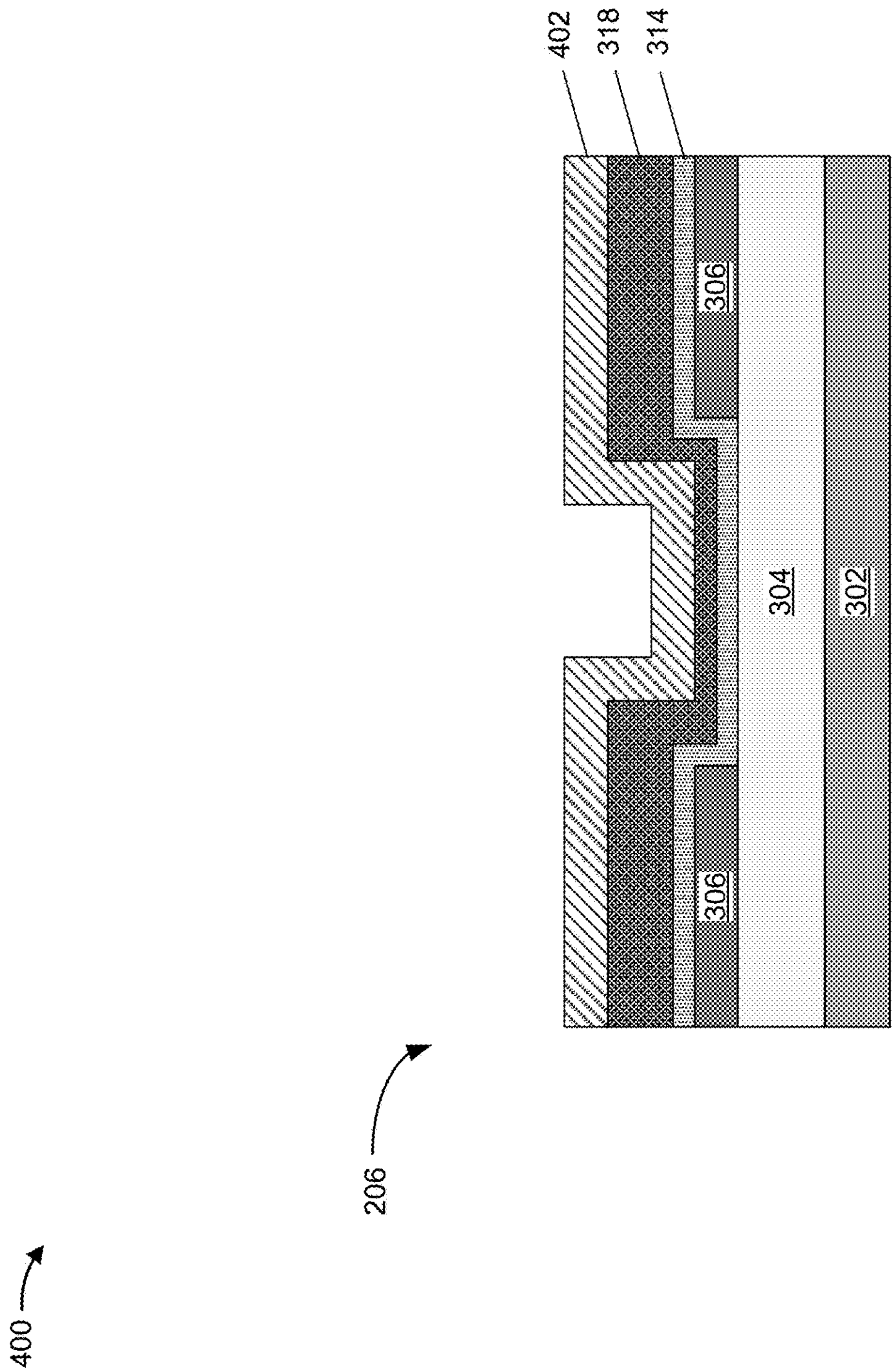


FIG. 4C

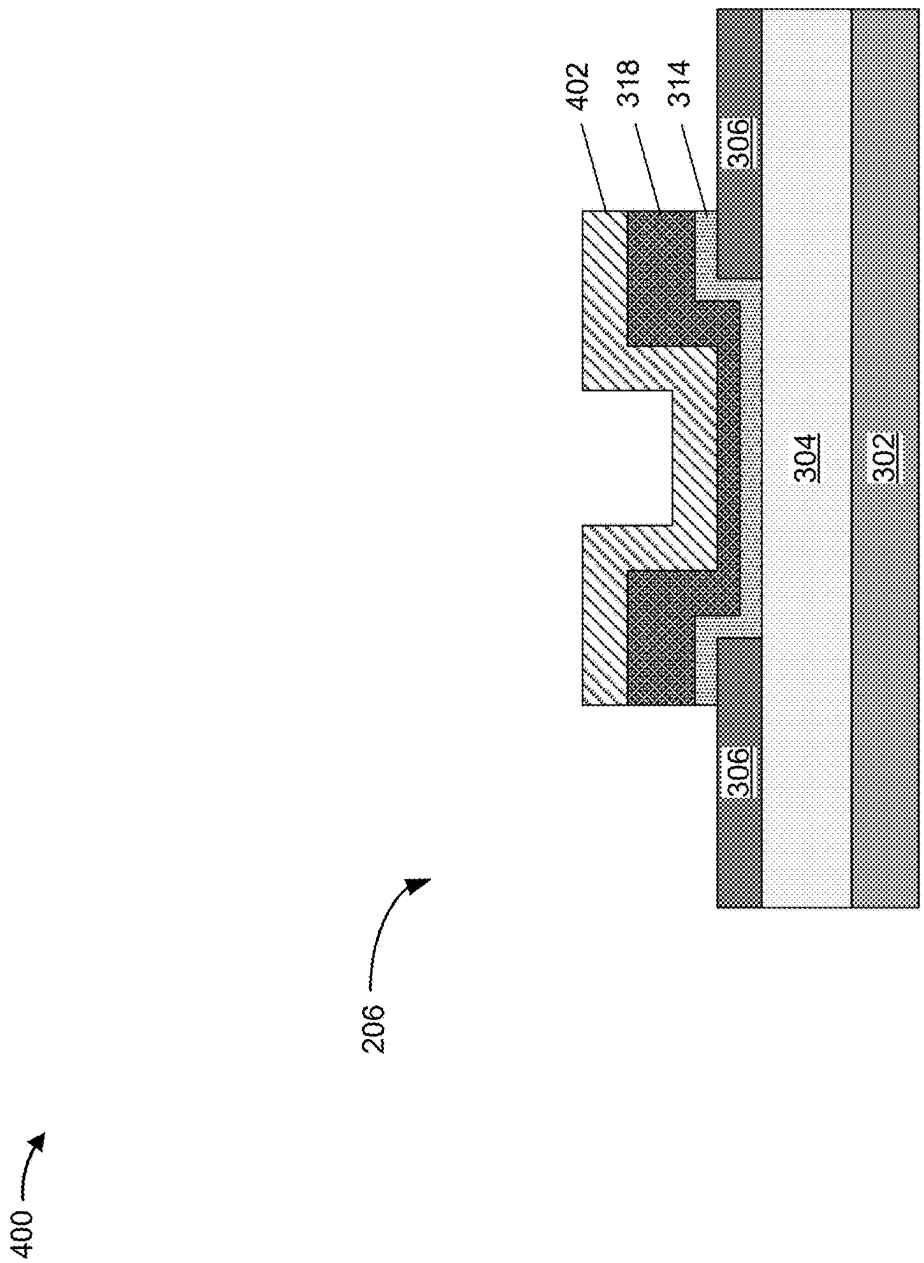


FIG. 4D

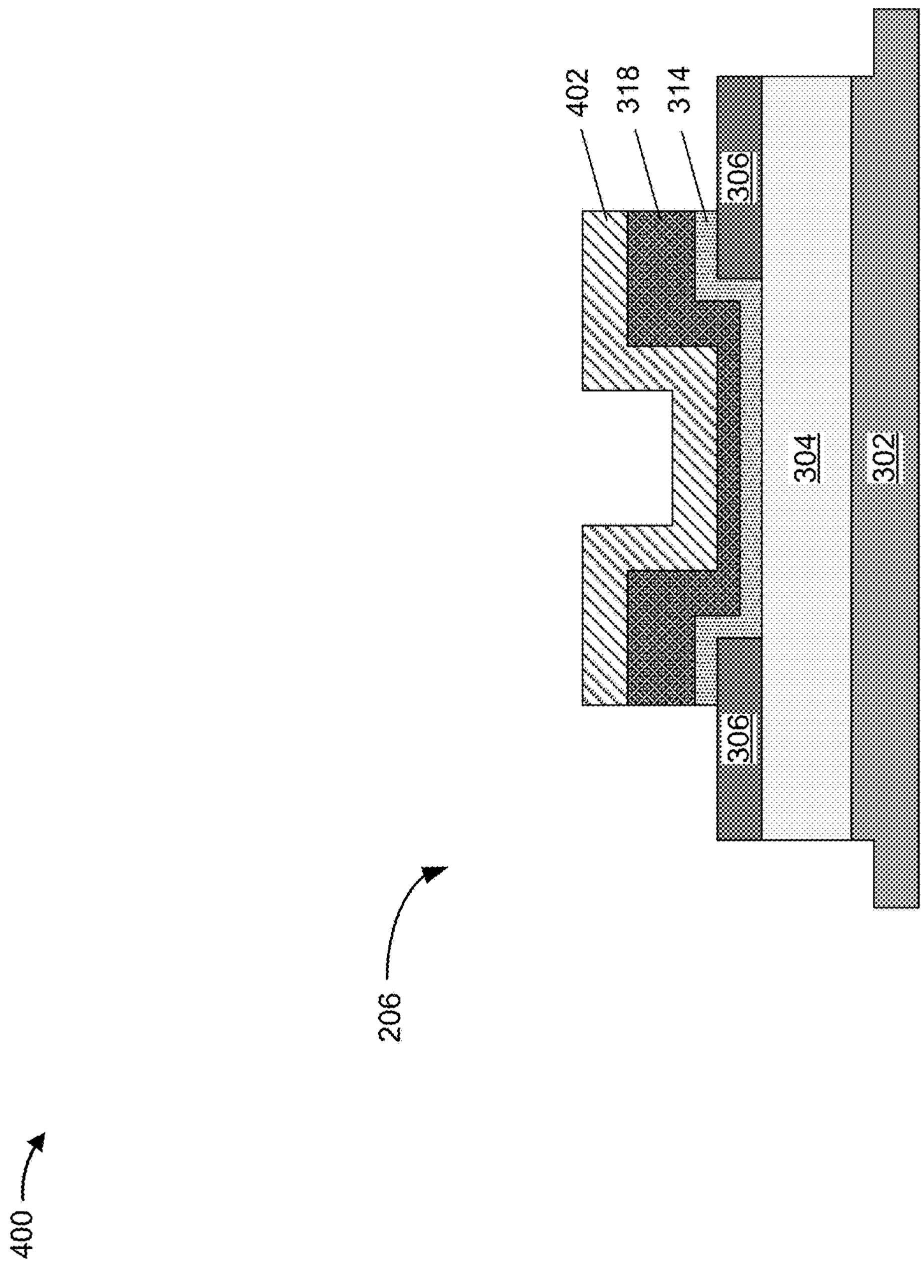


FIG. 4E

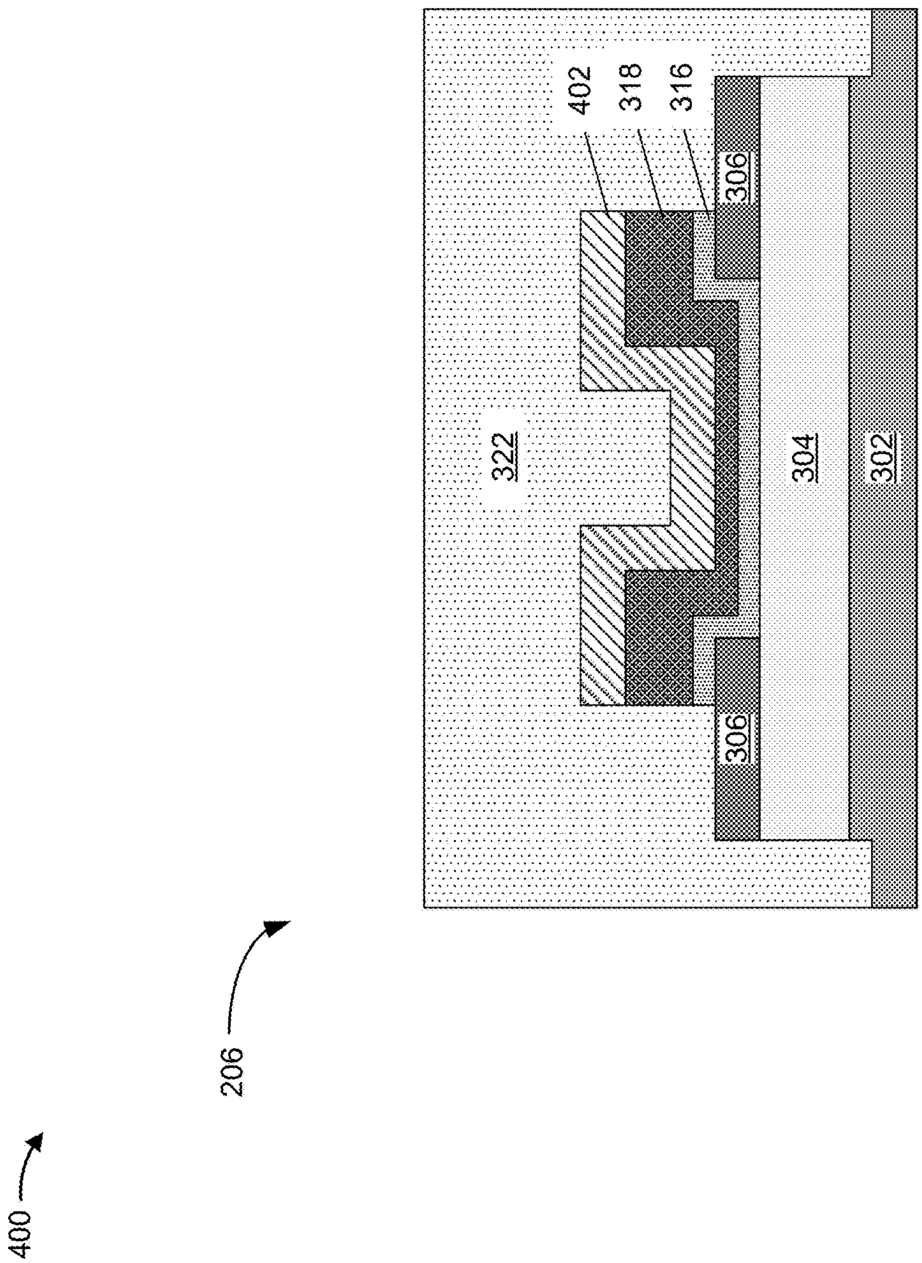


FIG. 4F

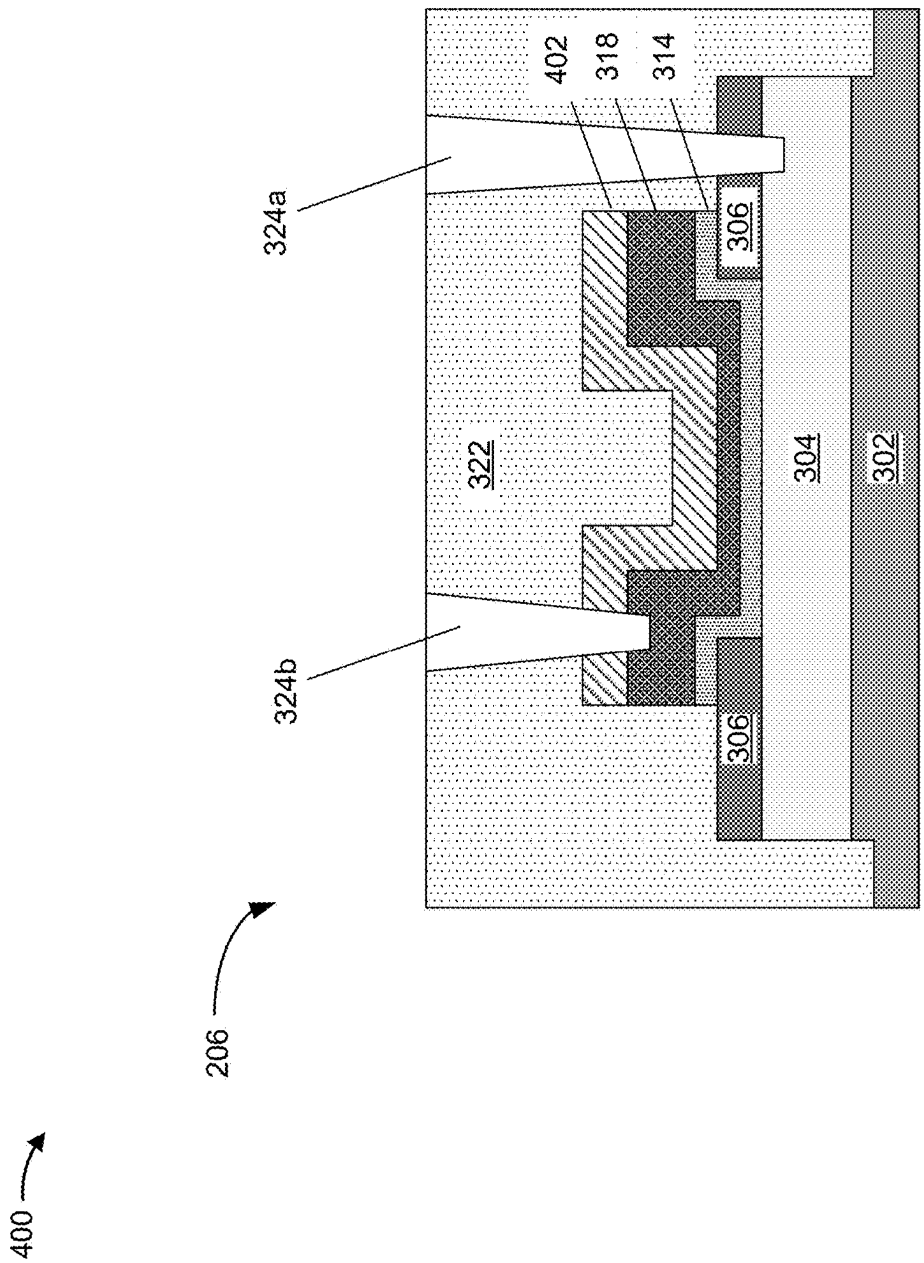


FIG. 4G

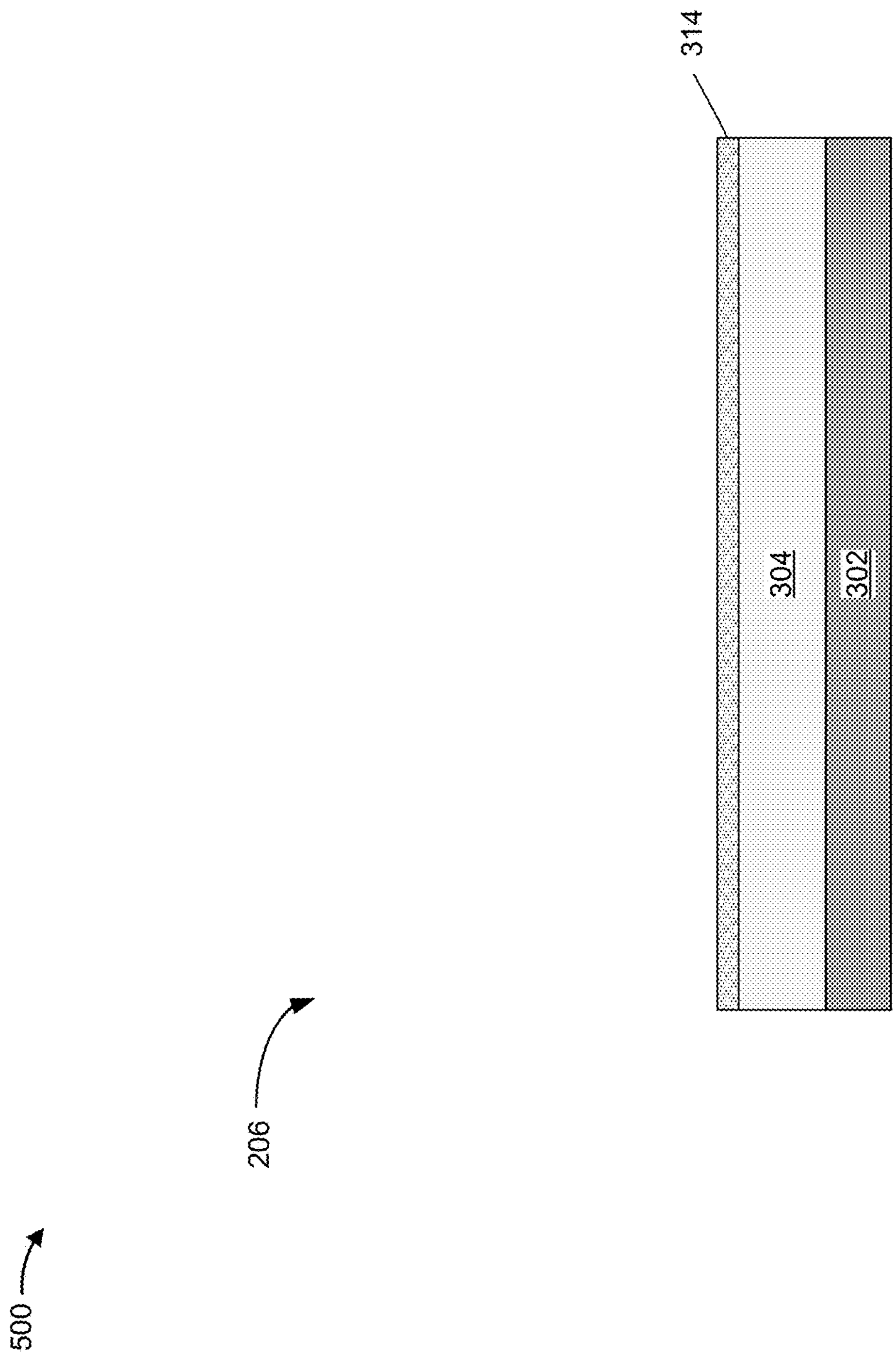


FIG. 5A

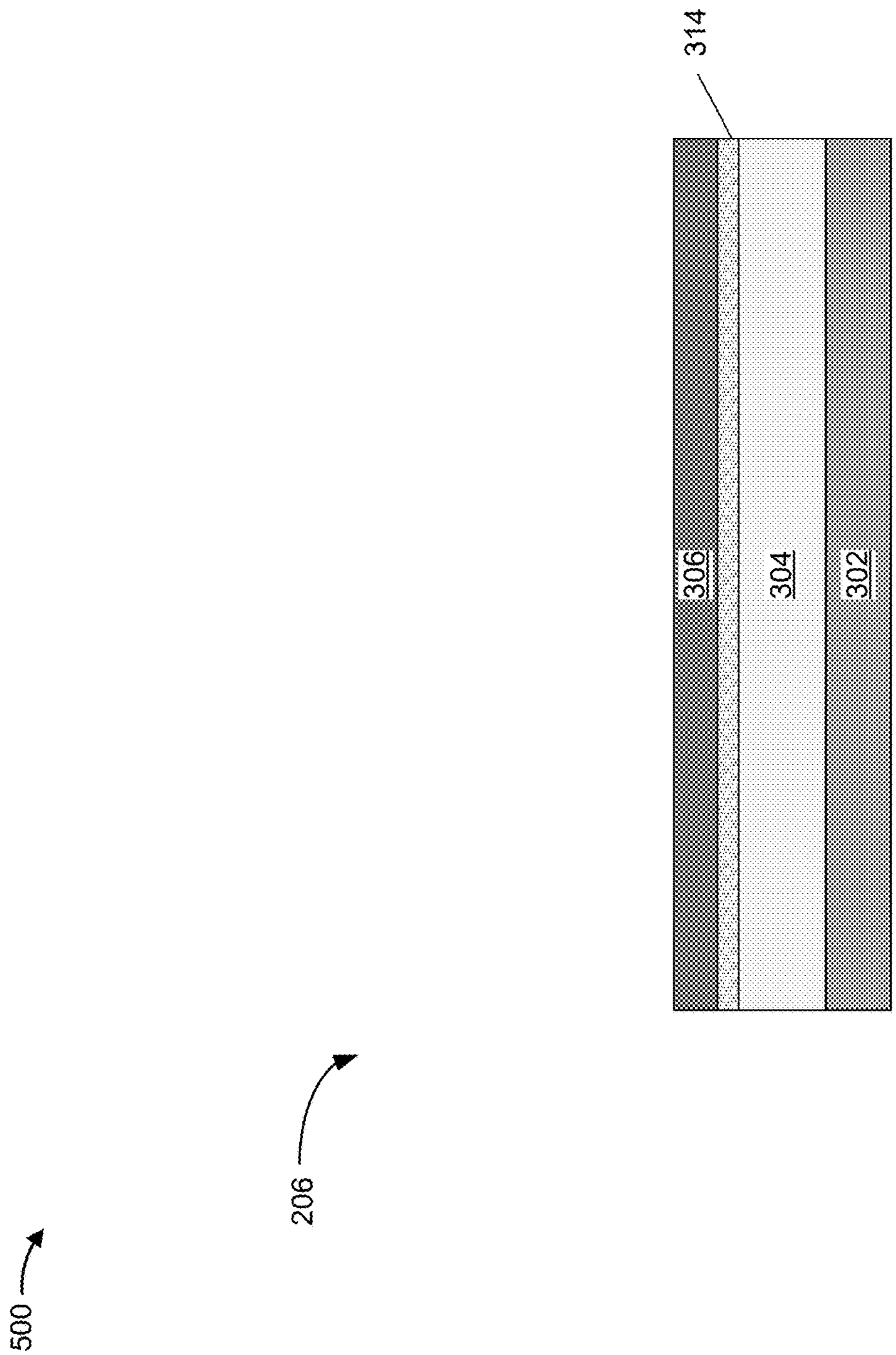


FIG. 5B

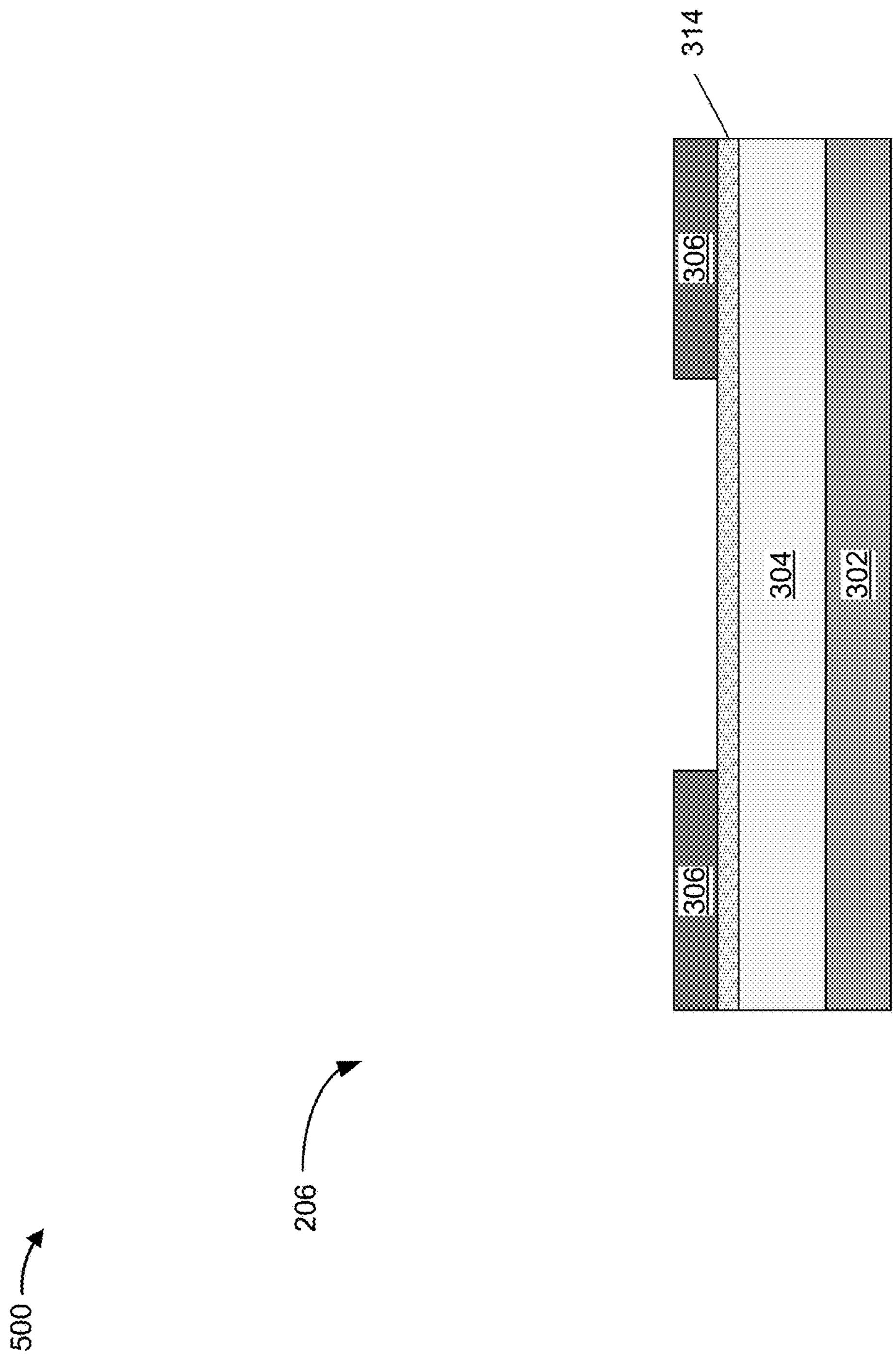


FIG. 5C

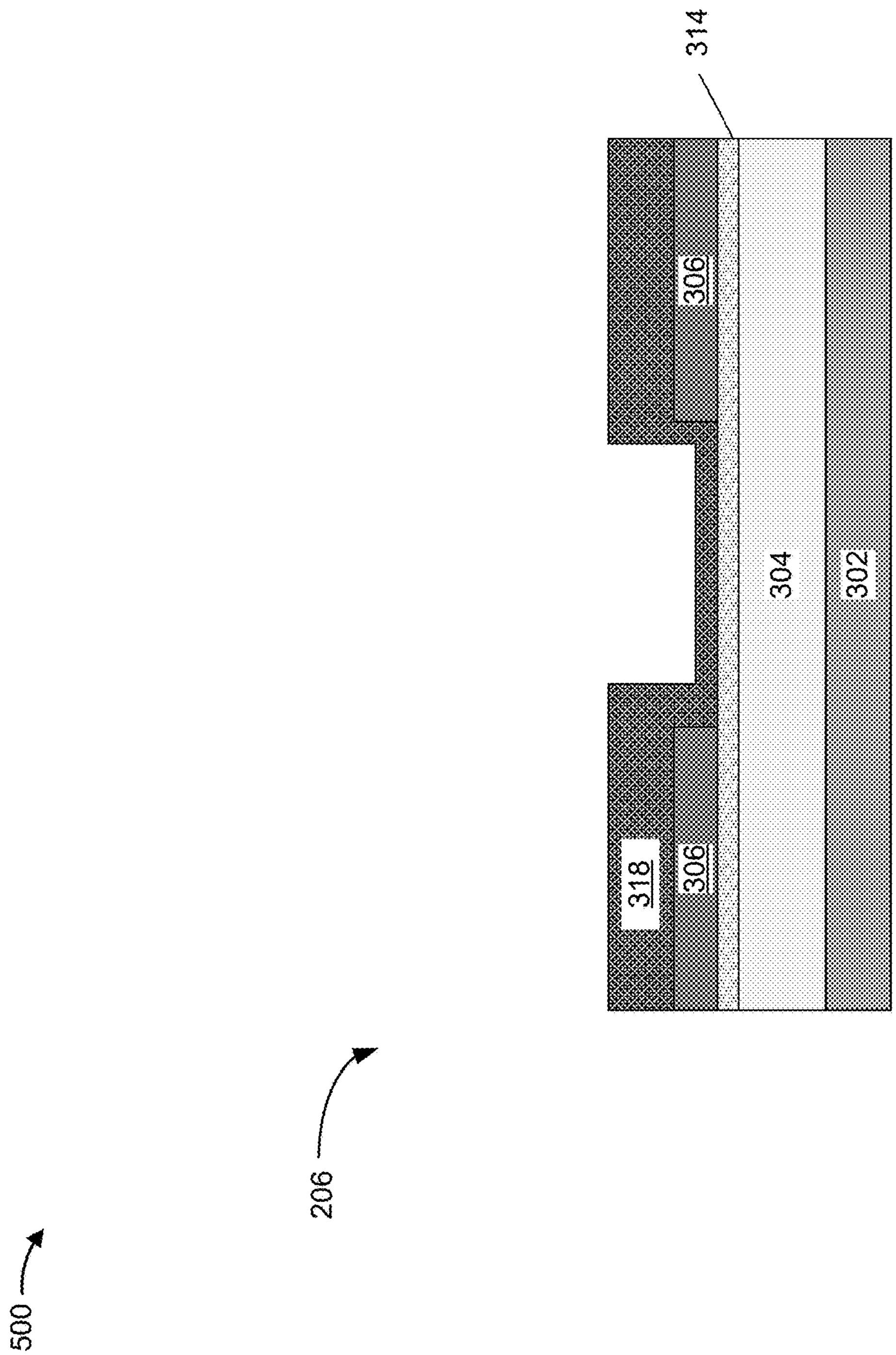


FIG. 5D

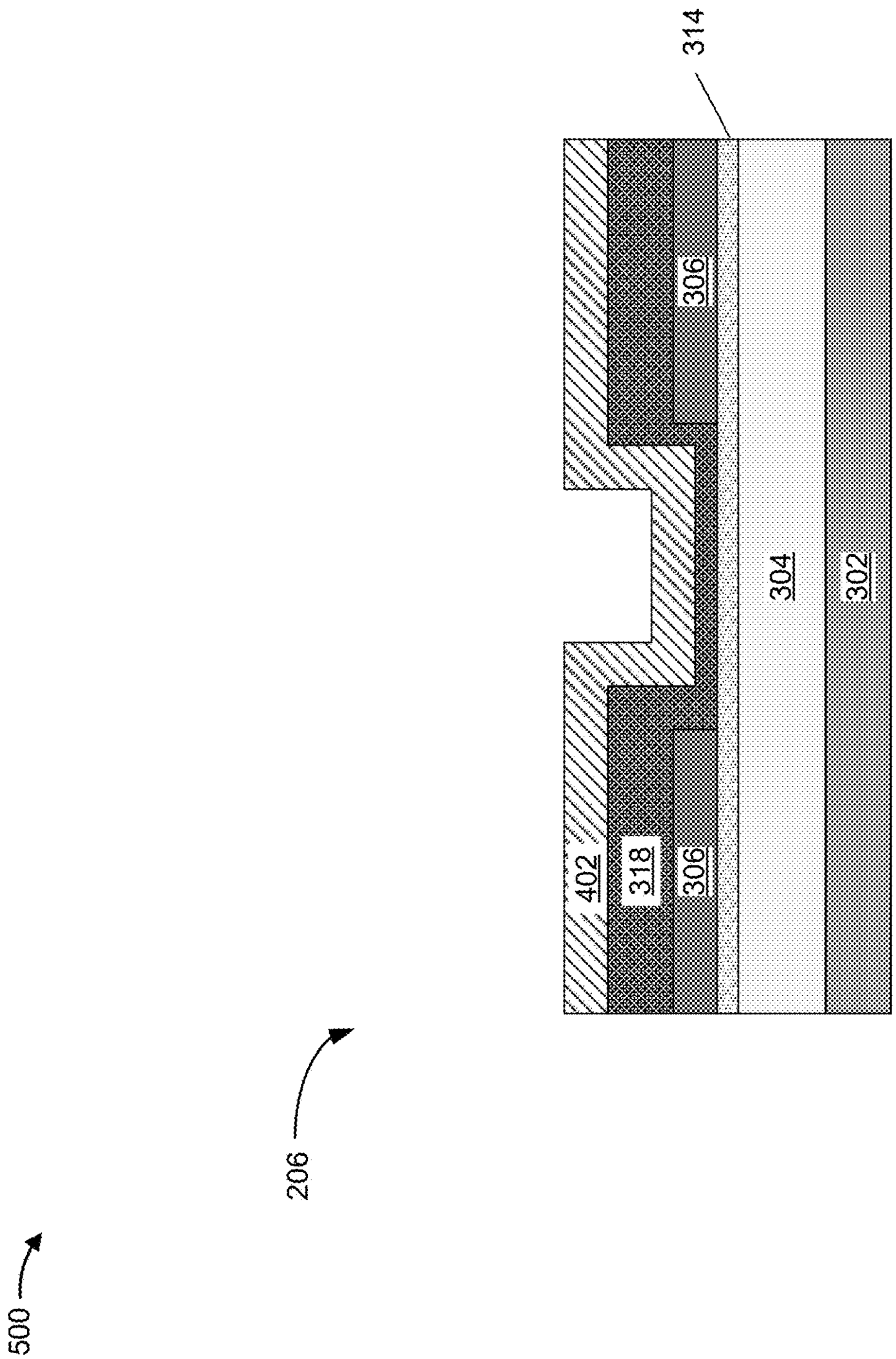


FIG. 5E

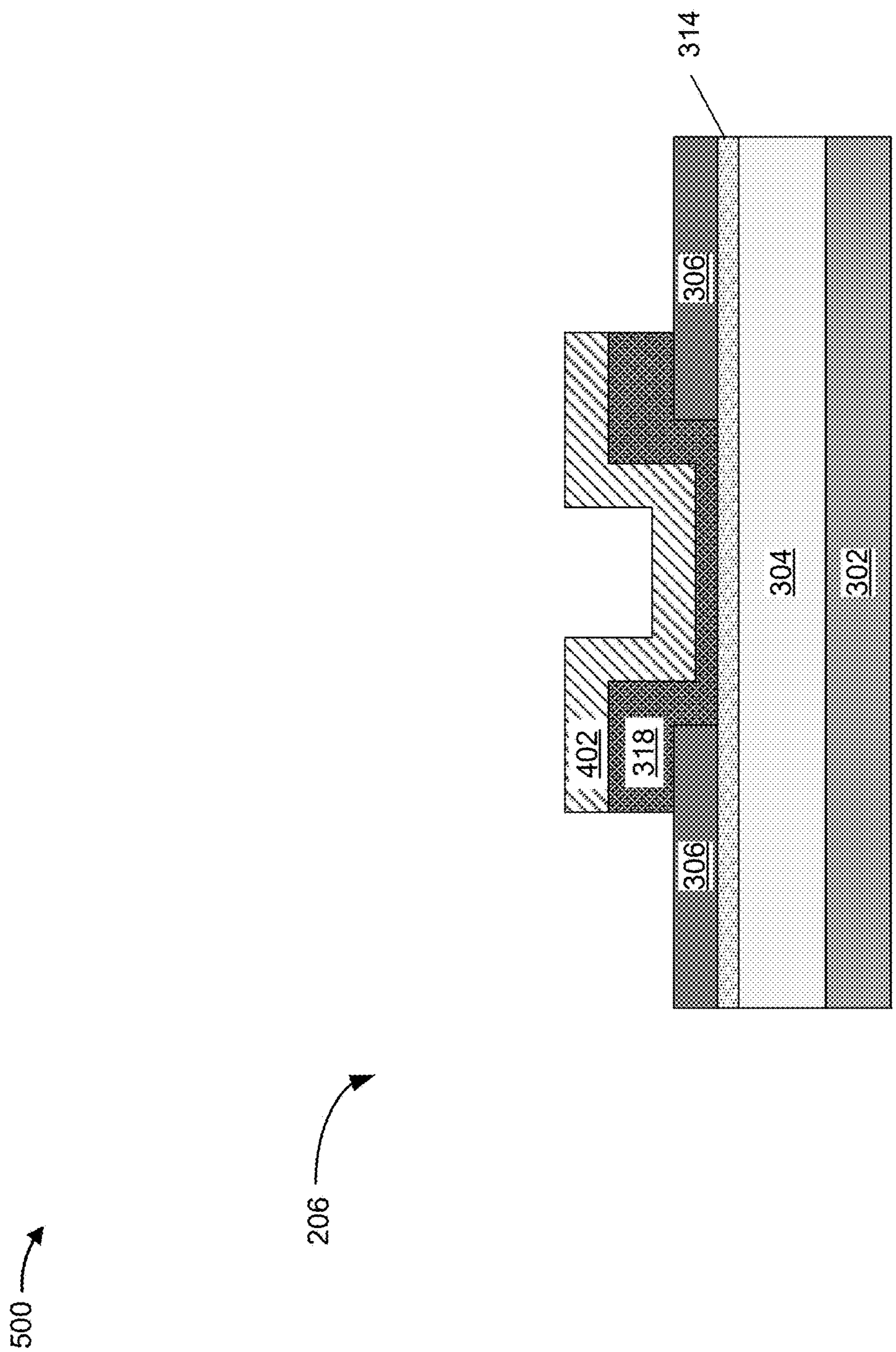


FIG. 5F

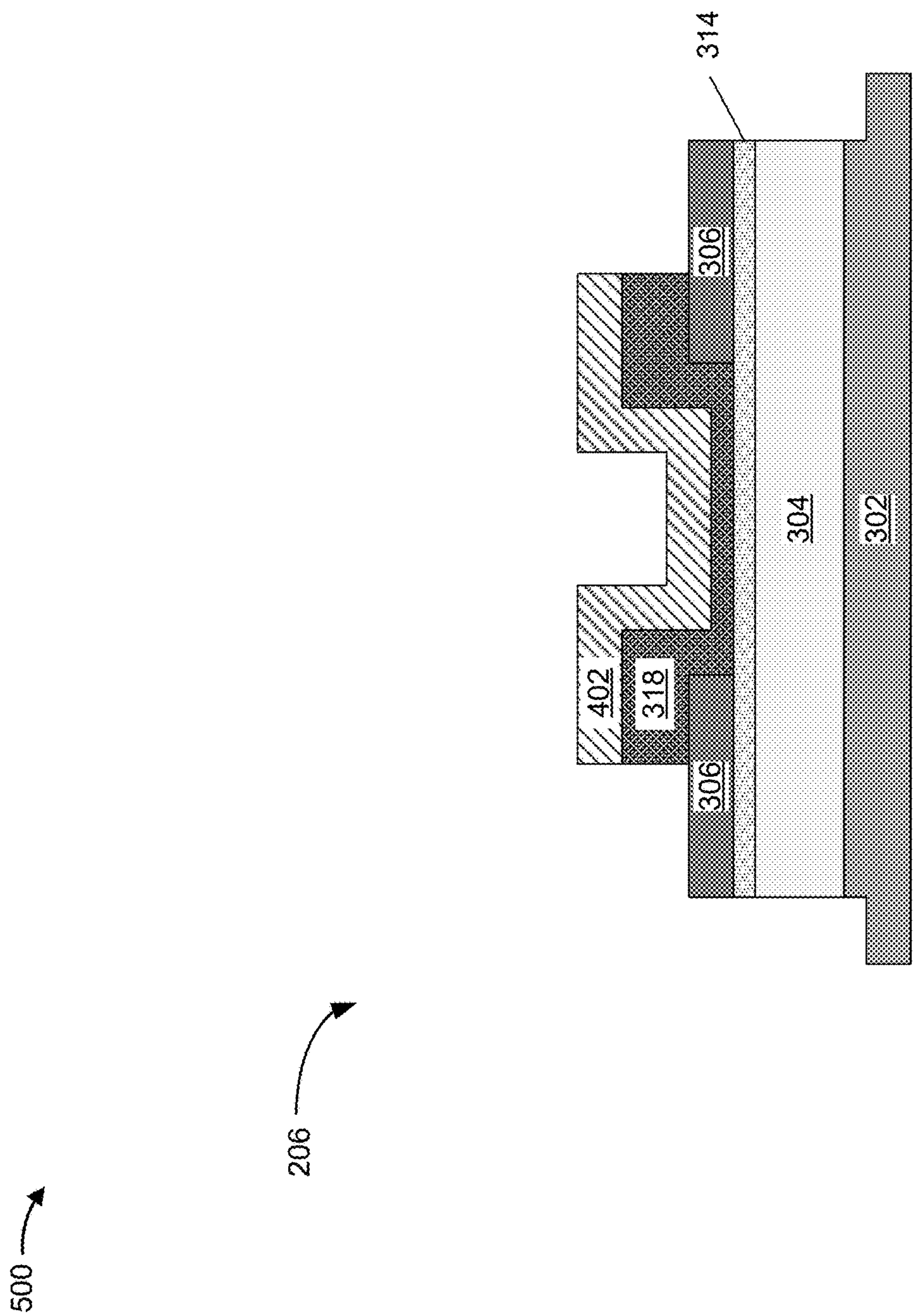


FIG. 5G

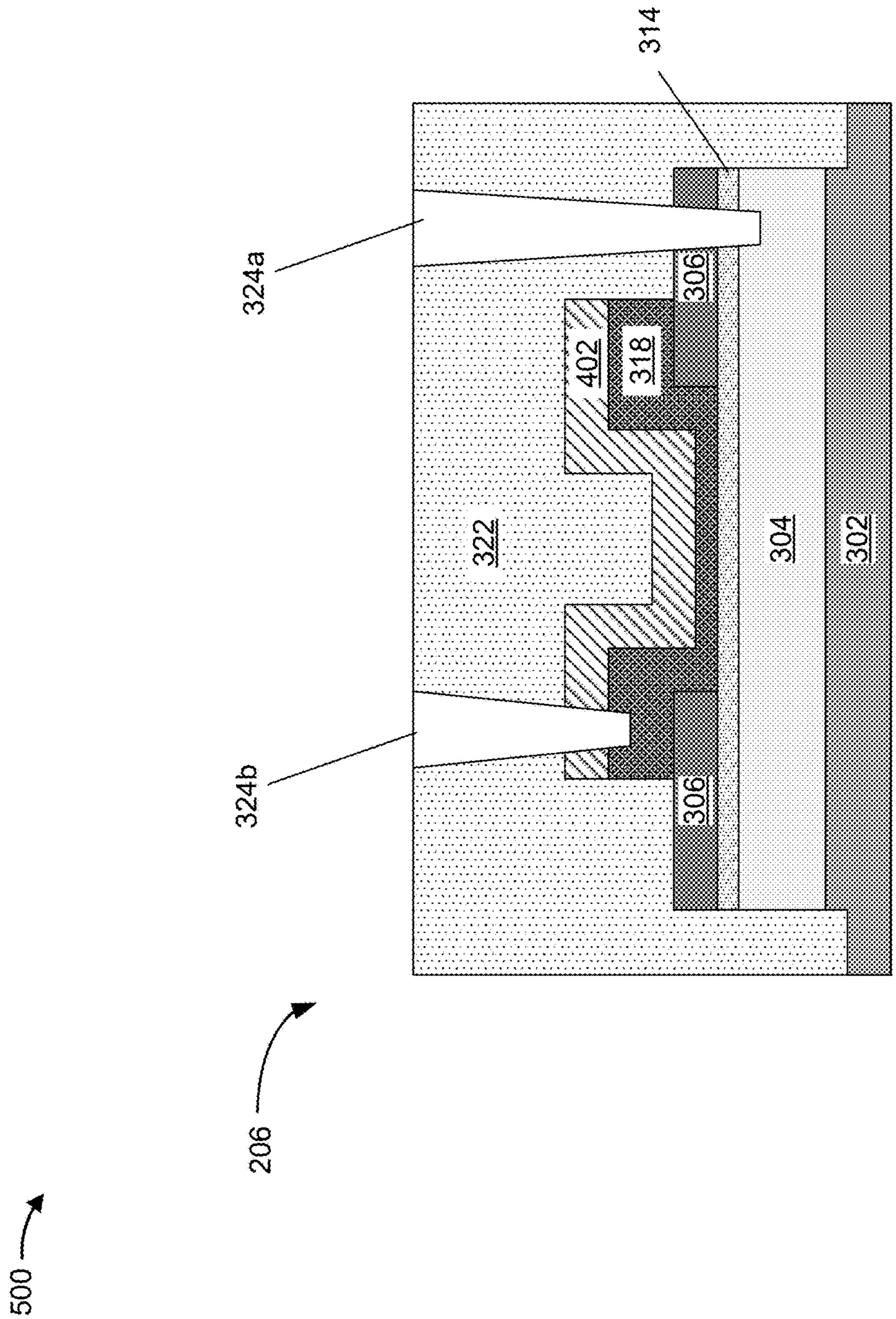


FIG. 5H

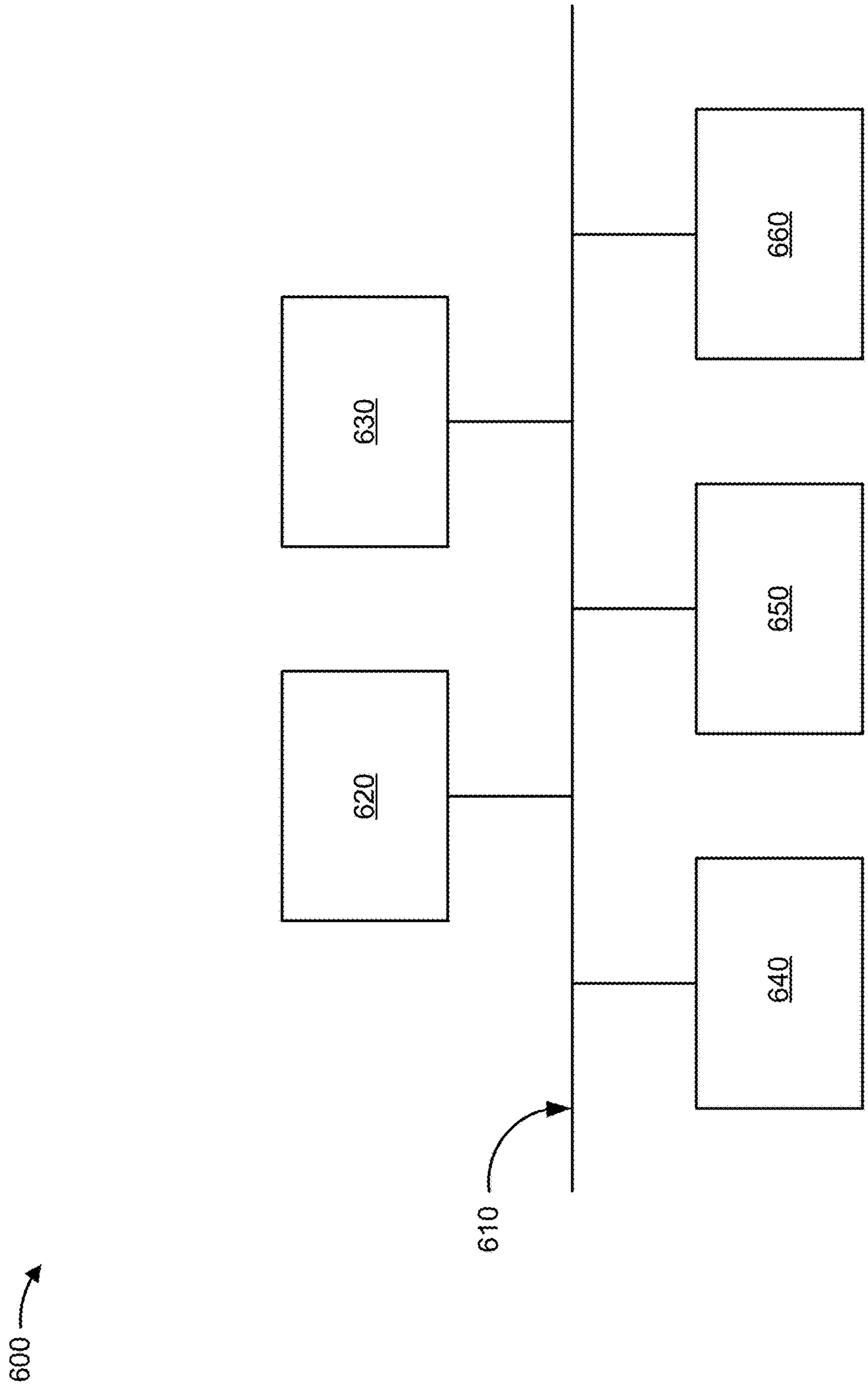


FIG. 6

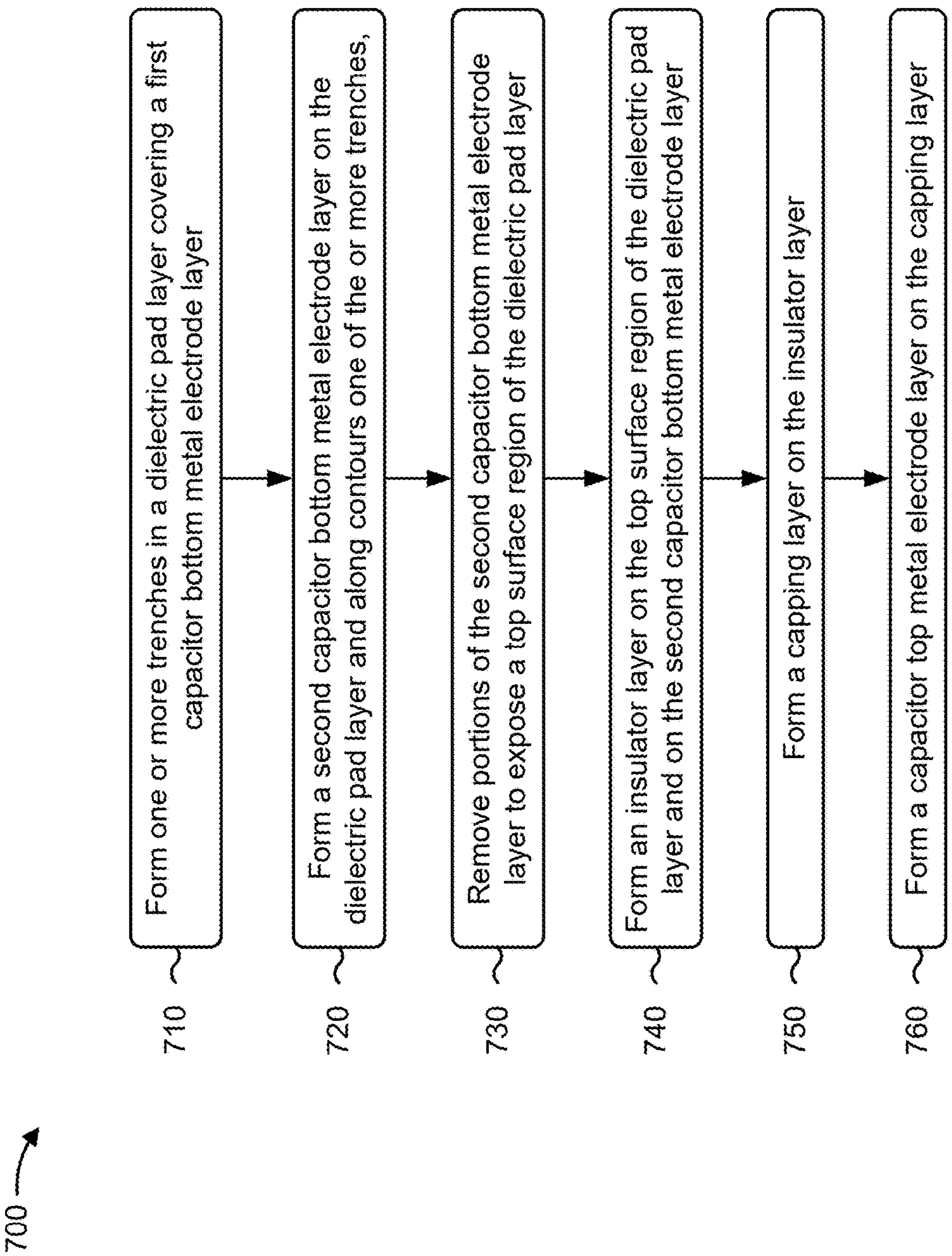

700
710
Form one or more trenches in a dielectric pad layer covering a first capacitor bottom metal electrode layer
720
Form a second capacitor bottom metal electrode layer on the dielectric pad layer and along contours one of the or more trenches,
730
Remove portions of the second capacitor bottom metal electrode layer to expose a top surface region of the dielectric pad layer
740
Form an insulator layer on the top surface region of the dielectric pad layer and on the second capacitor bottom metal electrode layer
750
Form a capping layer on the insulator layer
760
Form a capacitor top metal electrode layer on the capping layer

FIG. 7

METAL-INSULATOR-METAL CAPACITOR AND METHODS OF MANUFACTURING

RELATED APPLICATION

[0001] This application is a continuation of U.S. patent application Ser. No. 17/658,732, filed Apr. 11, 2022, which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] A metal-insulator-metal (MIM) device can be used as a capacitor in a semiconductor device. A MIM device includes two metal layers, with an insulator layer between the two metal layers.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. **1** is a diagram of an example environment in which systems and/or methods described herein may be implemented.

[0005] FIG. **2** is a diagram of an example implementation of a semiconductor device described herein.

[0006] FIGS. **3**A, **3**B, **3**C, **3**D, **3**E, **3**F, **3**G, **3**H, **3**I, and **3**J are diagrams of example implementations described herein.

[0007] FIGS. **4**A, **4**B, **4**C, **4**D, **4**E, **4**F, and **4**G are diagrams of example implementations described herein.

[0008] FIGS. **5**A, **5**B, **5**C, **5**D, **5**E, **5**F, **5**G, and **5**H are diagrams of example implementations described herein.

[0009] FIG. **6** is a diagram of example components of one or more devices of FIG. **1** described herein.

[0010] FIG. **7** is a flowchart of an example process associated with fabricating a metal-insulator-metal capacitor described herein.

DETAILED DESCRIPTION

[0011] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows includes embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0012] Further, spatially relative terms, such as "beneath," "below," "lower," "above," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0013] A metal-insulator-metal (MIM) device can be used as a capacitor in a semiconductor device. A MIM may include two metal layers, with an insulator layer between the two metal layers. Performance requirements of the MIM device in the semiconductor device may include a high capacitance and a low leakage current. The insulator layer may be between a capacitor bottom metal (CBM) electrode layer and a capacitor top metal (CTM) electrode layer. A thin insulator and/or HK dielectric may be applied to form a high capacitance MIM device.

[0014] During fabrication of the MIM device, a portion of the insulator layer may perform as an etch stop during formation of the CTM electrode layer. In such a case, and during etching of the CTM electrode layer, portions of the insulator layer may be thinned such that there is leakage or breakdown between the CBM electrode layer and the CTM electrode layer. Pre-emptively thickening the insulator layer to compensate for the leakage may cause an unetched portion of the insulator layer, between the CBM electrode layer and the CTM electrode layer, to have a thickness that results in a decreased capacitance of the MIM device.

[0015] Some implementations described herein provide techniques and apparatuses for forming a semiconductor device including a MIM capacitor. The MIM capacitor includes a dielectric pad layer having a portion between a CBM electrode layer and a portion of an insulator layer. The dielectric pad layer may preserve a thickness of the insulator layer to reduce a likelihood of a leakage between a CTM electrode layer and the CBM electrode layer. The dielectric pad layer may also enable a reduction in a thickness of the insulator layer to increase a capacitance of the MIM capacitor.

[0016] In this way, a performance of the MIM capacitor may be increased. Furthermore, in some implementations, the dielectric pad layer performs as an etch stop during formation of a layer stack including the dielectric pad layer, the CBM electrode layer, and the insulator layer to reduce use of an etch tool. In this way, a cost of the semiconductor device including the MIM capacitor may be decreased.

[0017] FIG. **1** is a diagram of an example environment **100** in which systems and/or methods described herein may be implemented. As shown in FIG. **1**, environment **100** may include a plurality of semiconductor processing tools **102**-**112** and a wafer/die transport tool **114**. The plurality of semiconductor processing tools **102**-**112** may include a deposition tool **102**, an exposure tool **104**, a developer tool **106**, an etch tool **108**, a planarization tool **110**, a plating tool **112**, and/or another type of semiconductor processing tool. The tools included in example environment **100** may be included in a semiconductor clean room, a semiconductor foundry, a semiconductor processing facility, and/or manufacturing facility, among other examples.

[0018] The deposition tool **102** is a semiconductor processing tool that includes a semiconductor processing chamber and one or more devices capable of depositing various types of materials onto a substrate. In some implementations, the deposition tool **102** includes a spin coating tool that is capable of depositing a photoresist layer on a substrate such as a wafer. In some implementations, the deposition tool **102** includes a chemical vapor deposition (CVD) tool such as a plasma-enhanced CVD (PECVD) tool, a high-density plasma CVD (HDP-CVD) tool, a sub-atmospheric CVD (SACVD) tool, a low-pressure CVD (LPCVD) tool, an atomic layer deposition (ALD) tool, a plasma-enhanced atomic layer deposition (PEALD) tool, or another type of CVD tool. In some implementations, the deposition tool **102** includes a physical vapor deposition (PVD) tool, such as a sputtering tool or another type of PVD tool. In some implementations, the deposition tool **102** includes an epitaxial tool that is configured to form layers and/or regions of a device by epitaxial growth. In some implementations, the example environment **100** includes a plurality of types of deposition tools **102**.

[0019] The exposure tool **104** is a semiconductor processing tool that is capable of exposing a photoresist layer to a radiation source, such as an ultraviolet light (UV) source (e.g., a deep UV light source, an extreme UV light (EUV) source, and/or the like), an x-ray source, an electron beam (e-beam) source, and/or the like. The exposure tool **104** may expose a photoresist layer to the radiation source to transfer a pattern from a photomask to the photoresist layer. The pattern may include one or more semiconductor device layer patterns for forming one or more semiconductor devices, may include a pattern for forming one or more structures of a semiconductor device, may include a pattern for etching various portions of a semiconductor device, and/or the like. In some implementations, the exposure tool **104** includes a scanner, a stepper, or a similar type of exposure tool.

[0020] The developer tool **106** is a semiconductor processing tool that is capable of developing a photoresist layer that has been exposed to a radiation source to develop a pattern transferred to the photoresist layer from the exposure tool **104**. In some implementations, the developer tool **106** develops a pattern by removing unexposed portions of a photoresist layer. In some implementations, the developer tool **106** develops a pattern by removing exposed portions of a photoresist layer. In some implementations, the developer tool **106** develops a pattern by dissolving exposed or unexposed portions of a photoresist layer through the use of a chemical developer.

[0021] The etch tool **108** is a semiconductor processing tool that is capable of etching various types of materials of a substrate, wafer, or semiconductor device. For example, the etch tool **108** may include a wet etch tool, a dry etch tool, and/or the like. In some implementations, the etch tool **108** includes a chamber that is filled with an etchant, and the substrate is placed in the chamber for a particular time period to remove particular amounts of one or more portions of the substrate. In some implementations, the etch tool **108** etches one or more portions of the substrate using a plasma etch or a plasma-assisted etch, which may involve using an ionized gas to isotropically or directionally etch the one or more portions. In some implementations, the etch tool **108** includes a plasma-based asher to remove a photoresist material.

[0022] The planarization tool **110** is a semiconductor processing tool that is capable of polishing or planarizing various layers of a wafer or semiconductor device. For example, a planarization tool **110** may include a chemical mechanical planarization (CMP) tool and/or another type of planarization tool that polishes or planarizes a layer or surface of deposited or plated material. The planarization tool **110** may polish or planarize a surface of a semiconductor device with a combination of chemical and mechanical forces (e.g., chemical etching and free abrasive polishing). The planarization tool **110** may utilize an abrasive and corrosive chemical slurry in conjunction with a polishing pad and retaining ring (e.g., typically of a greater diameter than the semiconductor device). The polishing pad and the semiconductor device may be pressed together by a dynamic polishing head and held in place by the retaining ring. The dynamic polishing head may rotate with different axes of rotation to remove material and even out any irregular topography of the semiconductor device, making the semiconductor device flat or planar.

[0023] The plating tool **112** is a semiconductor processing tool that is capable of plating a substrate (e.g., a wafer, a semiconductor device, and/or the like) or a portion thereof with one or more metals. For example, the plating tool **112** may include a copper electroplating device, an aluminum electroplating device, a nickel electroplating device, a tin electroplating device, a compound material or alloy (e.g., tin-silver, tin-lead, and/or the like) electroplating device, and/or an electroplating device for one or more other types of conductive materials, metals, and/or similar types of materials.

[0024] Wafer/die transport tool **114** includes a mobile robot, a robot arm, a tram or rail car, an overhead hoist transport (OHT) system, an automated materially handling system (AMHS), and/or another type of device that is configured to transport substrates and/or semiconductor devices between semiconductor processing tools **102-112**, that is configured to transport substrates and/or semiconductor devices between processing chambers of the same semiconductor processing tool, and/or that is configured to transport substrates and/or semiconductor devices to and from other locations such as a wafer rack, a storage room, and/or the like. In some implementations, wafer/die transport tool **114** may be a programmed device that is configured to travel a particular path and/or may operate semi-autonomously or autonomously. In some implementations, the environment **100** includes a plurality of wafer/die transport tools **114**.

[0025] For example, the wafer/die transport tool **114** may be included in a cluster tool or another type of tool that includes a plurality of processing chambers, and may be configured to transport substrates and/or semiconductor devices between the plurality of processing chambers, to transport substrates and/or semiconductor devices between a processing chamber and a buffer area, to transport substrates and/or semiconductor devices between a processing chamber and an interface tool such as an equipment front end module (EFEM), and/or to transport substrates and/or semiconductor devices between a processing chamber and a transport carrier (e.g., a front opening unified pod (FOUP)), among other examples. In some implementations, a wafer/die transport tool **114** may be included in a multi-chamber (or cluster) deposition tool **102**, which may include a pre-clean processing chamber (e.g., for cleaning or removing oxides, oxidation, and/or other types of contamination or byproducts from a substrate and/or semiconductor device) and a plurality of types of deposition processing chambers (e.g., processing chambers for depositing different types of materials, processing chambers for performing different types of deposition operations). In these implementations, the wafer/die transport tool **114** is configured to transport substrates and/or semiconductor devices between the processing chambers of the deposition tool **102** without breaking or removing a vacuum (or an at least partial vacuum) between the processing chambers and/or between processing operations in the deposition tool **102**, as described herein.

[0026] As described in connection with FIGS. **2-7** and elsewhere herein, the semiconductor processing tools **102-112** may perform a combination of operations to form a semiconductor device including a metal-insulator-metal (MIM) capacitor. As an example, the combination of operations includes forming one or more trench regions in a dielectric pad layer covering a first capacitor bottom metal (CBM) electrode layer. In some implementations, the one or more trench regions expose portions of the first CBM electrode layer. The combination of operations further includes forming a second CBM electrode layer on the dielectric pad layer along contours of the or more trench regions. In some implementations, portions of the second CBM electrode layer contact the portions of the first CBM electrode layer. The combination of operation also includes removing portions of the second CBM electrode layer to expose a top surface region of the dielectric pad layer, forming an insulator layer on the top surface region of the dielectric pad layer and on the second CBM electrode layer, and forming a capping layer on the insulator layer. The combination of operations also includes forming a capacitor top metal (CTM) electrode layer on the capping layer.

[0027] The number and arrangement of devices shown in FIG. **1** are provided as one or more examples. In practice, there may be additional devices, fewer devices, different devices, or differently arranged devices than those shown in FIG. **1**. Furthermore, two or more devices shown in FIG. **1** may be implemented within a single device, or a single device shown in FIG. **1** may be implemented as multiple, distributed devices. Additionally, or alternatively, a set of devices (e.g., one or more devices) of environment **100** may perform one or more functions described as being performed by another set of devices of environment **100**.

[0028] FIG. **2** is a diagram of an example implementation **200** of a semiconductor device **202** described herein. The semiconductor device **202** may include integrated circuitry **204** electrically coupled to a metal-insulator-metal (MIM) capacitor **206**.

[0029] As an example, the integrated circuitry **204** may correspond to memory circuitry and the MIM capacitor **206** may correspond to a memory cell. Additionally, or alternatively, the integrated circuitry **204** may correspond to imager circuitry and the MIM capacitor **206** may correspond to a pixel overflow capacitor. Additionally, or alternatively, the integrated circuitry **204** may correspond to transistor circuitry and the MIM capacitor **206** may correspond to a coupling capacitor. However, other types of integrated circuitry (e.g., the integrated circuitry **204**) are within the scope of the present disclosure.

[0030] As described in connection with FIGS. **3A-7**, and elsewhere herein, the semiconductor device **202** may include the integrated circuitry **204** and the MIM capacitor **206**. The MIM capacitor **206** includes an insulator layer between a CTM electrode layer and a CBM electrode layer and a dielectric pad layer including a first portion between the CTM electrode layer and the CBM electrode layer. The MIM capacitor **206** includes a first interconnect access structure passing through a second portion of the dielectric pad layer to electrically connect to the CBM electrode layer and a second interconnect access structure passing through the CTM electrode layer to make physical contact with a capping layer. In some implementations, the second interconnect access structure passing through the CTM electrode layer causes the second interconnect access structure to electrically connect to the CTM electrode layer above another portion of the dielectric pad layer. In some implementations, the first interconnect access structure and the second interconnect access structure electrically couple the MIM capacitor **206** to the integrated circuitry **204**.

[0031] Additionally, or alternatively, the MIM capacitor **206** includes a CTM electrode layer and a CBM electrode layer. The MIM capacitor **206** includes an insulator layer between the CTM electrode layer and the CBM electrode layer. The MIM capacitor **206** includes a dielectric pad layer having a portion between the CTM electrode layer and the CBM electrode layer. The MIM capacitor **206** further includes an interconnect access structure passing through another portion of the dielectric pad layer to electrically connect to the CBM electrode layer.

[0032] As indicated above, FIG. **2** is provided as an example. Other examples may differ from what is described with regard to FIG. **2**.

[0033] FIGS. **3A-3J** are diagrams of an example implementation **300** described herein. As described in connection with FIGS. **3A-3J**, one or more of the semiconductor processing tools **102-112** may perform a series of operations to form the MIM capacitor **206**.

[0034] FIG. **3A** shows a cross-sectional view of a portion of the MIM capacitor **206**, including a dielectric layer **302**, a CBM electrode layer **304***a* (e.g., a first CBM electrode layer), and a dielectric pad layer **306**. The dielectric layer **302** may include, as an example, a tantalum nitride (TaN) material, a silicon oxide (SiO_x) material, a silicate glass material, a silicon oxycarbide (SiOC) material, a silicon nitride (Si_xN_y) material, and/or the like.

[0035] The deposition tool **102** may deposit, on the dielectric layer **302**, a conductive metal capable of carrying an electric charge such as an aluminum copper (AlCu) material, a titanium (Ti) material, a titanium nitride (TiN) material, a tantalum (Ta) material, a tantalum nitride (TaN) material, or a tungsten (W) material, among other examples, to form the CBM electrode layer **304***a*. The CBM electrode layer **304***a* may have a thickness D1 that is included in a range of approximately 5 nanometers to approximately 500 nanometers. However, other values and ranges for the thickness D1 are within the scope of the present disclosure. The deposition tool **102** may deposit the conductive metal to form the CBM electrode layer **304***a* using a CVD technique, a PVD technique, an ALD technique, a plating technique described above in connection with FIG. **1**, and/or another deposition technique.

[0036] The deposition tool **102** may deposit, on the CBM electrode layer **304***a*, one or more materials to form the dielectric pad layer **306**. The one or more materials may include a silicon nitride (SiN) material, a silicon dioxide (SiO_2) material, a silicon oxynitride (SiON) material, or a silicon oxy carbonitride (SiOCN) material, among other examples. Additionally, or alternatively, the one or more materials may include a high-k dielectric material such as a hafnium oxide (HfO_2) material, an aluminum oxide (Al_2O_3) material, a zirconium dioxide (ZrO_2) material, or a titanium dioxide (TiO_2) material, among other examples. However, other materials for the dielectric pad layer **306** are within the scope of the present disclosure. The deposition tool **102** may deposit the one or more materials to form the dielectric pad layer **306** using a CVD technique, a PVD technique, an ALD technique, a plating technique described above in connection with FIG. **1**, and/or another deposition technique.

[0037] A thickness D**2** of the dielectric pad layer **306** may vary based on a configuration of the MIM capacitor **206**. As an example, the thickness D**2** may be included in a range of approximately 20 nanometers to approximately 2000 nanometers. If the thickness D**2** is less than approximately 20 nanometers, gains in capacitance (e.g., capacitor area) from a trench configuration would be minimized (and capacitance would be decreased). Additionally, or alternatively, use of the dielectric pad layer **306** as an etch stop in subsequent manufacturing steps would be compromised, and leakage or breakdown may occur within the MIM capacitor **206**. If the thickness is greater than approximately 2000 nanometers, a depth of trench regions in the trench configuration may impede subsequent deposition of an insulator layer, a capping layer, and/or a CTM electrode layer along contours of the trench regions, leading to defects and a decrease in yield of the semiconductor device **202** including the dielectric pad layer **306**.

[0038] FIG. **3**B shows a cross-sectional view including one or more trench regions **308** being formed in the dielectric pad layer **306**. In some implementations, the etch tool **108** removes portions of the dielectric pad layer **306** to form the one or more trench regions **308**. The etch tool **108** may remove the portions of the dielectric layer **306** using a plasma etch technique, a wet chemical etch technique, and/or another type of etch technique described above in connection with FIG. **1**, and/or another etch technique. Forming the one or more trench regions **308** may expose portions of the CBM electrode layer **304***a*.

[0039] FIG. **3**C shows a cross-sectional view including a CBM electrode layer **304***b* (e.g., a second CBM electrode layer) being formed on the dielectric pad layer **306** and along contours of the one or more trench regions **308**. In some implementations, the deposition tool **102** deposits a conductive metal material capable of carrying an electric charge such as an aluminum copper (AlCu) material, a titanium (Ti) material, a titanium nitride (TiN) material, a tantalum (Ta) material, a tantalum nitride (TaN) material, or a tungsten (W) material, among other examples, to form the CBM electrode layer **304***b*. However, other conductive metal materials are within the scope of the present disclosure. The CBM electrode layer **304***b* may electrically connect with the CBM electrode layer **304***a* at the bottom of the one or more trench regions **308**. The deposition tool **102** may deposit the conductive metal material to form the CBM electrode layer **304***b* using a CVD technique, a PVD technique, an ALD technique, a plating technique described above in connection with FIG. **1**, and/or another deposition technique.

[0040] FIG. **3**D shows a cross-sectional view including portions of the CBM electrode layer **304***b* being removed to expose one or more top surface regions **312** of the dielectric pad layer **306**. In some implementations the etch tool **108** removes the portions of the CBM electrode layer **304***b* to expose the one or more top surface regions **312**. The etch tool **108** may remove the portions of the CBM electrode layer **304***b* using a plasma etch technique, a wet chemical etch technique, and/or another type of etch technique described above in connection with FIG. **1**, and/or another etch technique.

[0041] FIG. **3**E shows a cross-sectional view including an insulator layer **314** being formed on the one or more top surface regions **312** and on the CBM electrode layer **304***b*. The insulator layer **314**, as shown, includes a shape that conforms to the one or more trench regions **308**. In some implementations, the deposition tool **102** deposits one or more or more materials to form the insulator layer **314**. The one or more materials may include a silicon nitride (SiN) material, a silicon dioxide (SiO_2) material, or a silicon oxy carbonitride (SiOCN) material, among other examples. Additionally, or alternatively, the one or more materials may include a high-k dielectric material such as a hafnium oxide (HfO_2) material, an aluminum oxide (Al_2O_3) material, a zirconium dioxide (ZrO_2) material, or a titanium dioxide (TiO_2) material, among other examples. However, other materials for the insulator layer **314** are within the scope of the present disclosure. The deposition tool **102** may deposit the one or more materials to form the insulator layer **314** using a CVD technique, a PVD technique, an ALD technique, a plating technique described above in connection with FIG. **1**, and/or another deposition technique.

[0042] The insulator layer **314** may include a thickness D**3**. As an example, the thickness D**3** may be included in a range of approximately 1 nanometer to approximately 100 nanometers. If the thickness D**3** is less than approximately 1 nanometer, uniformity of deposition of the insulator layer **314** throughout the one or more trench regions **308** might be impeded, leading to voids and/or defects between the dielectric pad layer **306** and the insulator layer **314** to decrease a yield of the semiconductor device **202** including the insulator layer **314**. If the thickness D**3** is greater than approximately 100 nanometers, a capacitance of the MIM capacitor **206** might be reduced to decrease a performance of the semiconductor device **202** including the insulator layer **314**. However, other values and ranges for the thickness D**3** are within the scope of the present disclosure.

[0043] In some implementations, and as shown in the cross-sectional view of FIG. **3**E, a capping layer **316** is formed on the insulator layer **314**. For example, the deposition tool **102** may deposit a tungsten (W) material, among other examples, to form the capping layer **316**. This structure of including the insulator layer **314** between the capping layer **316** and the dielectric pad layer **306**, as shown in FIG. **3**E, may preserve a thickness of the insulator layer **314** during subsequent manufacturing steps to reduce a likelihood of leakage (e.g., leakage between the CBM electrode layers **304***a*, **304***b* and a subsequently formed CTM electrode layer) within the MIM capacitor **206**. Additionally, the thickness D**3** of the insulator layer **314** may be tightly controlled to achieve a desired capacitance performance of the MIM capacitor **206**. The deposition tool **102** may form the insulator layer **314** using a CVD technique, a PVD technique, an ALD technique, a plating technique described above in connection with FIG. **1**, and/or another deposition technique.

[0044] FIG. **3**F shows a cross-sectional view including formation of a CTM electrode layer **318** on the capping layer **316**. In some implementations, the deposition tool **102** deposits a conductive metal capable of carrying an electric charge such as an aluminum copper (AlCu) material, a titanium (Ti) material, a titanium nitride (TiN) material, a tantalum (Ta) material, a tantalum nitride (TaN) material, or a tungsten (W) material, among other examples, to form the CTM electrode layer **318**. The deposition tool **102** may deposit the conductive metal to form the CTM electrode layer **318** using a CVD technique, a PVD technique, an ALD technique, a plating technique described above in connection with FIG. **1**, and/or another deposition technique.

[0045] FIG. **3**G shows a cross-sectional view including removal of portions of the CTM electrode layer **318**, the capping layer **316**, the insulator layer **314**, and the dielectric pad layer **306** to expose one or more recessed surface regions **320** of the dielectric pad layer **306**. For example, the etch tool **108** may perform one or more etching operations to remove the portions of the CTM electrode layer **318**, the capping layer **316**, and/or the insulator layer **314** to expose the one or more surface regions **320** of the dielectric pad layer **306**. The etch tool **108** may perform the one or more etching operations using a plasma etch technique, a wet chemical etch technique, and/or another type of etch technique described above in connection with FIG. **1**, and/or another etch technique. After removal of the portions of the CTM electrode layer **318**, the capping layer **316**, the insulator layer **314**, and/or the dielectric pad layer **306**, edges of the insulator layer **314** and the capping layer **316** are aligned (e.g., "self-aligned") to edges of the CTM electrode layer **318**.

[0046] As shown in FIG. **3**G, the dielectric pad layer **306** may perform as an etch stop during the removal of the portions of the CTM electrode layer **318**, the capping layer **316**, and the insulator layer **314**. In this way, utilization of the etch tool **108** may be increased (e.g., eliminating multiple etch cycles and/or etch recipes) to reduce a cost of the semiconductor device **202** including the MIM capacitor **206**.

[0047] In some implementations, and as shown in the cross-sectional view of FIG. **3**H, other portions of the dielectric pad layer **306** and portions of the CBM electrode layer **304***a* may be removed. For example, the etch tool **108** may perform one or more etching operations to remove the other portions of the dielectric pad layer **306** and the portions of the CBM electrode layer **304***a* to create one or more regions for deposition of an inter-layer dielectric (ILD) layer included the MIM capacitor **206**. The etch tool **108** may perform the one or more etching operations using a plasma etch technique, a wet chemical etch technique, and/or another type of etch technique described above in connection with FIG. **1**, and/or another etch technique.

[0048] FIG. **3**I shows a cross-sectional view including deposition of an ILD layer **322** included in the MIM capacitor **206**. For example, the deposition tool **102** may deposit a silicon nitride (SiN) material, a silicon dioxide (SiO_2) material, or a silicon oxy carbonitride (SiOCN) material, among other examples, to form the ILD layer **322**. However, other materials for the ILD layer **322** are within the scope of the present disclosure. The ILD layer **322** may provide electrical isolation between one or more layers of the MIM capacitor **206**. The deposition tool **102** may deposit the ILD layer **322** using a CVD technique, a PVD technique, an ALD technique, a plating technique described above in connection with FIG. **1**, and/or another deposition technique.

[0049] FIG. **3**J shows a cross-sectional view including formation of interconnect access structures **324** (e.g., vertical interconnect access structures, or "vias") through the ILD layer **322** (shown as interconnect access structure **324***a* and interconnect access structure **324***b* in FIG. **3**J). For example, the etch tool **108** may form recesses (e.g., cavities) within the ILD layer **322** and the deposition tool **102** may deposit an electrically conductive material (e.g., copper (Cu) or tungsten (W), among other examples) within the recesses to form the interconnect access structures **324**. The etch tool **108** may form the recesses using a plasma etch technique, a wet chemical etch technique, and/or another type of etch technique described above in connection with FIG. **1**, and/or another etch technique. The deposition tool may deposit the electrically conductive material to form the interconnect access structures using a CVD technique, a PVD technique, an ALD technique, a plating technique described above in connection with FIG. **1**, and/or another deposition technique. The interconnect access structures **324** may provide an electrically conductive path to electrically couple the MIM capacitor **206** to integrated circuitry of the semiconductor device **202** (e.g., the integrated circuitry **204**).

[0050] As shown in FIG. **3**J, the MIM capacitor **206** includes the CTM electrode layer **318** and the CBM electrode layer **304***a*. The MIM capacitor **206** further includes the insulator layer **314** between the CTM electrode layer **318** and the CBM electrode layer **304***a*. The MIM capacitor **206** includes the dielectric pad layer **306**, including a portion **306***a* between the CTM electrode layer **318** and the CBM electrode layer **304***a*. As shown in FIG. **3**J, the portion **306***a* of the dielectric pad layer **306** is below a portion of the insulator layer **314**.

[0051] FIG. **3**J shows additional aspects of the MIM capacitor **206**, including the capping layer **316** between the CTM electrode layer **318** and the insulator layer **314**. FIG. **3**J further shows edges of the insulator layer **314** aligned to edges of the CTM electrode layer **318**.

[0052] The MIM capacitor **206** includes the interconnect access structure **324***a* (e.g., corresponding to a first interconnect access structure) passing through another portion **306***b* of the dielectric pad layer **306** to electrically connect to the CBM electrode layer **304***a*. The MIM capacitor **206** includes the interconnect access structure **324***b* (e.g., corresponding to a second interconnect access structure) passing through the CTM electrode layer **318** to make physical contact with the capping layer **316**. In some implementations, and as shown, the interconnect access structure **324***b* passing through the CTM electrode layer **318** causes the interconnect access structure **324***b* to electrically connect to the CTM electrode layer **318** above another portion **306***c* of the dielectric pad layer **306**.

[0053] The number and arrangement of layers as shown in FIGS. **3**A-**3**J are provided as one or more examples. In practice, and as described in connection with FIGS. **4**A-**5**H, there may be additional layers, different layers, or differently arranged layers than those shown in FIGS. **3**A-**3**J. Furthermore, different combinations of layers, arrangements of layers, or materials included in the MIM capacitor **206** may be implemented within a single semiconductor device **202**.

[0054] FIGS. **4**A-**4**G are diagrams of an example implementation **400** described herein. The example implementation **400** may use similar fabrication techniques and materials as described in connection with the example implementation **300** of FIGS. **3**A-**3**J to form the MIM capacitor **206**. However, in contrast to the example implementation **300** of FIGS. **3**A-**3**J, multiple trench regions are absent from MIM capacitor **206**.

[0055] Cross-sectional views of FIGS. **4**A and **4**B show formation of the dielectric layer **302**, the CBM electrode layer **304**, and the dielectric pad layer **306**. In contrast to the example implementation of FIGS. **3**A-**3**J, the implementation **400** includes a single CBM electrode layer **304** (implementation **300** of FIGS. **3**A-**3**J include two CBM electrode layers, e.g., the CBM electrode layer **304***a* and the CBM electrode layer **304***b*).

[0056] Cross-sectional views of FIGS. **4**C-**4**E show formation of the insulator layer **314**, the CTM electrode layer **318**, and a dielectric layer **402**. The dielectric layer **402** may include, for example, a silicon nitride (SiN) material, a silicon dioxide (SiO_2) material, or a silicon oxy carbonitride (SiOCN), among other examples. However, other materials for the dielectric layer **402** are within the scope of the present disclosure.

[0057] As shown in FIG. **4**D, the dielectric pad layer **306** may perform as an etch stop during removal of portions of the dielectric layer **402**, CTM electrode layer **318**, and the insulator layer **314**. In this way, a utilization of the etch tool **108** may be increased (e.g., eliminating multiple etch cycles and/or etch recipes) to reduce a cost of the semiconductor device **202** including the MIM capacitor **206**.

[0058] In some implementations, and as shown in FIG. **4**E, this structure of including the insulator layer **314** between the CTM electrode layer **318** and the dielectric pad layer **306** may preserve a thickness of the insulator layer **314** during subsequent manufacturing steps to reduce a likelihood of leakage between the CTM electrode layer **318** and the CBM electrode layer **304** within the MIM capacitor **206**. Additionally, a thickness of the insulator layer **314** may be tightly controlled to achieve a desired capacitance performance of the MIM capacitor **206**.

[0059] FIGS. **4**F and **4**G show cross-sectional views that include formation of the ILD layer **322** and the interconnect access structures **324**. The ILD layer **322** may provide electrical isolation between one or more layers of the MIM capacitor **206**. The interconnect access structures **324** (e.g., interconnect access structure **324***a* and interconnect access structure **324***b*) may provide an electrically conductive path to electrically couple the MIM capacitor **206** to integrated circuitry of the semiconductor device **202** (e.g., the integrated circuitry **204**).

[0060] Similar to the MIM capacitor **206** formed using techniques described in connection with FIGS. **3**A-**3**J, a portion of the dielectric pad layer **306** between the CTM electrode layer **318** and the CBM electrode layer **304** is below a portion of the insulator layer **314**.

[0061] As indicated above, FIGS. **4**A-**4**G are provided as examples. Other examples may differ from what is described with regard to FIGS. **4**A-**4**G.

[0062] FIGS. **5**A-**5**H are diagrams of an example implementation **500** described herein. The example implementation **500** may use similar fabrication techniques and materials as described in connection with the example implementation **300** of FIGS. **3**A-**3**J to form the MIM capacitor **206**. However, in contrast to the example implementation **300** of FIGS. **3**A-**3**J, and/or the example implementation **400** of FIGS. **4**A-**4**G, implementation **500** includes the insulator layer **314** being formed prior to the dielectric pad layer **306**.

[0063] Cross-sectional views of FIGS. **5**A-**5**C show formation of the dielectric layer **302**, the CBM electrode layer **304**, the insulator layer **314**, and the dielectric pad layer **306**. In particular, and as shown in FIG. **5**A, the insulator layer **314** is formed prior to the dielectric pad layer **306**.

[0064] FIGS. **5**E-**5**G show formation of the CTM electrode layer **318** and the dielectric layer **402** over the CTM electrode layer **318**. In some implementations, and as shown in FIG. **5**E, this structure of including the insulator layer **314** between the dielectric pad layer **306** and the CBM electrode layer **304** may preserve a thickness of the insulator layer **314** during subsequent manufacturing steps to reduce a likelihood of leakage between the CTM electrode layer **318** and the CBM electrode layer **304** within the MIM capacitor **206**. Additionally, a thickness of the insulator layer **314** may be tightly controlled to achieve a desired capacitance performance of the MIM capacitor **206**.

[0065] As shown in FIG. **5**F, the dielectric pad layer **306** may perform as an etch stop during the removal of the portions of the dielectric layer **402** and the CTM electrode layer **318**. In this way, utilization of the etch tool **108** may be increased (e.g., eliminating multiple etch cycles and/or etch recipes) to reduce a cost of the semiconductor device **202** including the MIM capacitor **206**.

[0066] FIG. **5**H shows cross-sectional views that include formation of the ILD layer **322** and the interconnect access structures **324** (shown as interconnect access structure **324***a* and interconnect access structure **324***b*). The ILD layer **322** may provide electrical isolation between one or more layers of the MIM capacitor **206**. The interconnect access structures **324** may provide an electrically conductive path to electrically couple the MIM capacitor **206** to integrated circuitry of the semiconductor device **202** (e.g., the integrated circuitry **204**).

[0067] In contrast to the MIM capacitor **206** formed using techniques described in connection with FIGS. **3**A-**3**J and/or FIGS. **4**A-**4**G, and as shown in FIG. **5**H, a portion of the dielectric pad layer **306** between the CTM electrode layer **318** and the CBM electrode layer **304** is above a portion of the insulator layer **314**.

[0068] As indicated above, FIGS. **5**A-**5**H are provided as examples. Other examples may differ from what is described with regard to FIGS. **5**A-**5**H.

[0069] FIG. **6** is a diagram of example components of one or more devices of FIG. **1** described herein. In some implementations, one or more of the semiconductor processing tools **102**-**112** include one or more devices **600** and/or one or more components of device **600**. As shown in FIG. **6**, device **600** may include a bus **610**, a processor **620**, a memory **630**, an input component **640**, an output component **650**, and a communication component **660**.

[0070] Bus **610** includes one or more components that enable wired and/or wireless communication among the components of device **600**. Bus **610** may couple together two or more components of FIG. **6**, such as via operative coupling, communicative coupling, electronic coupling, and/or electric coupling. Processor **620** includes a central processing unit, a graphics processing unit, a microprocessor, a controller, a microcontroller, a digital signal processor, a field-programmable gate array, an application-specific integrated circuit, and/or another type of processing component. Processor **620** is implemented in hardware, firmware, or a combination of hardware and software. In some implementations, processor **620** includes one or more processors capable of being programmed to perform one or more operations or processes described elsewhere herein.

[0071] Memory **630** includes volatile and/or nonvolatile memory. For example, memory **630** may include random access memory (RAM), read only memory (ROM), a hard disk drive, and/or another type of memory (e.g., a flash memory, a magnetic memory, and/or an optical memory).

Memory **630** may include internal memory (e.g., RAM, ROM, or a hard disk drive) and/or removable memory (e.g., removable via a universal serial bus connection). Memory **630** may be a non-transitory computer-readable medium. Memory **630** stores information, instructions, and/or software (e.g., one or more software applications) related to the operation of device **600**. In some implementations, memory **630** includes one or more memories that are coupled to one or more processors (e.g., processor **620**), such as via bus **610**.

[0072] Input component **640** enables device **600** to receive input, such as user input and/or sensed input. For example, input component **640** may include a touch screen, a keyboard, a keypad, a mouse, a button, a microphone, a switch, a sensor, a global positioning system sensor, an accelerometer, a gyroscope, and/or an actuator. Output component **650** enables device **600** to provide output, such as via a display, a speaker, and/or a light-emitting diode. Communication component **660** enables device **600** to communicate with other devices via a wired connection and/or a wireless connection. For example, communication component **660** may include a receiver, a transmitter, a transceiver, a modem, a network interface card, and/or an antenna.

[0073] Device **600** may perform one or more operations or processes described herein. For example, a non-transitory computer-readable medium (e.g., memory **630**) may store a set of instructions (e.g., one or more instructions or code) for execution by processor **620**. Processor **620** may execute the set of instructions to perform one or more operations or processes described herein. In some implementations, execution of the set of instructions, by one or more processors **620**, causes the one or more processors **620** and/or the device **600** to perform one or more operations or processes described herein. In some implementations, hardwired circuitry is used instead of or in combination with the instructions to perform one or more operations or processes described herein. Additionally, or alternatively, processor **620** may be configured to perform one or more operations or processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

[0074] The number and arrangement of components shown in FIG. **6** are provided as an example. Device **600** may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. **6**. Additionally, or alternatively, a set of components (e.g., one or more components) of device **600** may perform one or more functions described as being performed by another set of components of device **600**.

[0075] FIG. **7** is a flowchart of an example process associated with fabricating a metal-insulator-metal capacitor described herein. In some implementations, one or more process blocks of FIG. **7** are performed by one or more semiconductor processing tools (e.g., one or more of the semiconductor processing tools **102-112**). Additionally, or alternatively, one or more process blocks of FIG. **7** may be performed by one or more components of device **600**, such as processor **620**, memory **630**, input component **640**, output component **650**, and/or communication component **660**.

[0076] As shown in FIG. **7**, process **700** may include forming one or more trench regions in a dielectric pad layer covering a first CBM electrode layer (block **710**). For example, one or more of the semiconductor processing tools **102-112**, such as the etch tool **108**, may form one or more trench regions **308** in a dielectric pad layer **306** covering a first CBM electrode layer **304***a*, as described above. In some implementations, the one or more trench regions **308** expose portions of the first CBM electrode layer **304***a*.

[0077] As further shown in FIG. **7**, process **700** may include forming a second CBM electrode layer on the dielectric pad layer and along contours of the one or more trench regions (block **720**). For example, one or more of the semiconductor processing tools **102-112**, such as the deposition tool **102**, may form a second CBM electrode layer **304***b* on the dielectric pad layer and along contours of the or more trench regions **308**, as described above. In some implementations, portions of the second CBM electrode layer **304***b* contact the portions of the first CBM electrode layer **304***a*.

[0078] As further shown in FIG. **7**, process **700** may include removing portions of the second CBM electrode layer to expose a top surface region of the dielectric pad layer (block **730**). For example, one or more of the semiconductor processing tools **102-112**, such as the etch tool **108**, may remove portions of the second CBM electrode layer **304***b* to expose a top surface region **312** of the dielectric pad layer **306**, as described above.

[0079] As further shown in FIG. **7**, process **700** may include forming an insulator layer on the top surface region of the dielectric pad layer and on the second CBM electrode layer (block **740**). For example, one or more of the semiconductor processing tools **102-112**, such as the deposition tool **102**, may form an insulator layer **314** on the top surface region **312** of the dielectric pad layer **306** and on the second CBM electrode layer **304***b*, as described above.

[0080] As further shown in FIG. **7**, process **700** may include forming a capping layer on the insulator layer (block **750**). For example, one or more of the semiconductor processing tools **102-112**, such as the deposition tool **102**, may form a capping layer **316** on the insulator layer, as described above.

[0081] As further shown in FIG. **7**, process **700** may include forming a CTM electrode layer on the capping layer (block **760**). For example, one or more of the semiconductor processing tools **102-112**, such as the deposition tool **102**, may form a CTM electrode layer **318** on the capping layer **316**, as described above.

[0082] Process **700** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

[0083] In a first implementation, process **700** includes forming the dielectric pad layer **306** by depositing a layer of a silicon nitride material, a layer of a silicon dioxide material, a layer of a silicon oxynitride material, or a layer of a silicon oxy carbonitride material.

[0084] In a second implementation, alone or in combination with the first implementation, process **700** includes forming the dielectric pad layer **306** by depositing a layer of a hafnium oxide material, a layer of a zirconium dioxide material, a layer of a titanium oxide material, or a layer of an aluminum oxide material.

[0085] In a third implementation, alone or in combination with one or more of the first and second implementations, process **700** includes removing portions of the CTM electrode layer **318**, the capping layer **316**, the insulator layer **314**, and the dielectric pad layer **306** to expose a recessed surface region **320** of the dielectric pad layer **306**, and forming an inter-layer dielectric layer **322** on the CTM electrode layer **318** and the recessed surface region **320** of the dielectric pad layer **306**.

[0086] In a fourth implementation, alone or in combination with one or more of the first through third implementations, process **700** includes forming, through the inter-layer dielectric layer and **320** through the dielectric pad layer **306** at the recessed surface region **320**, an interconnect access structure **324***a* to electrically connect to the first CBM electrode layer **304***a*.

[0087] In a fifth implementation, alone or in combination with one or more of the first through fourth implementations, process **700** includes forming, through the inter-layer dielectric layer **322** and through the CTM electrode layer **318**, an interconnect access structure **324***b* to physically contact the capping layer **316** and to electrically connect to the CTM electrode layer **318**.

[0088] Although FIG. **7** shows example blocks of process **700**, in some implementations, process **700** includes additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **7**. Additionally, or alternatively, two or more of the blocks of process **700** may be performed in parallel.

[0089] Some implementations described herein provide techniques and apparatuses for forming a semiconductor device including a MIM capacitor. The MIM capacitor includes a dielectric pad layer having a portion between a CBM electrode layer and a portion of an insulator layer. The dielectric pad layer may preserve a thickness of the insulator layer to reduce a likelihood of a leakage or breakdown between a CTM electrode layer and the CBM electrode layer. The dielectric pad layer may also enable a reduction in a thickness of the insulator layer to increase a capacitance of the MIM capacitor.

[0090] In this way, a performance of the MIM capacitor may be increased. Furthermore, in some implementations, the dielectric pad layer performs as an etch stop during formation of a layer stack including the dielectric pad layer, the CBM electrode layer, and the insulator layer to reduce use of an etch tool. In this way, a cost of the semiconductor device including the MIM capacitor may be decreased.

[0091] As described in greater detail above, some implementations described herein provide a device. The device includes a CTM electrode layer. The device includes a CBM electrode layer. The device includes an insulator layer between CTM electrode layer and the CBM electrode layer. The device includes a dielectric pad layer including a portion between the CTM electrode layer and the CBM electrode layer. The device includes an interconnect access structure passing through another portion of the dielectric pad layer to electrically connect to the CBM electrode layer.

[0092] As described in greater detail above, some implementations described herein provide a method. The method includes forming one or more trench regions in a dielectric pad layer covering a first CBM electrode layer, where the one or more trench regions expose portions of the first CBM electrode layer. The method includes forming a second CBM electrode layer on the dielectric pad layer and along contours of the or more trench regions, where portions of the second CBM electrode layer contact the portions of the first CBM electrode layer. The method includes removing portions of the second CBM electrode layer to expose a top surface region of the dielectric pad layer. The method includes forming an insulator layer on the top surface region of the dielectric pad layer and on the second CBM electrode layer. The method includes forming a capping layer on the insulator layer. The method includes forming a CTM electrode layer on the capping layer.

[0093] As described in greater detail above, some implementations described herein provide a semiconductor device. The semiconductor device includes integrated circuitry. The semiconductor device includes a metal-insulator-metal capacitor having an insulator layer between a CTM electrode layer and a CBM electrode layer. The metal-insulator-metal capacitor includes a dielectric pad layer including a first portion between the CTM electrode layer and the CBM electrode layer. The metal-insulator-metal capacitor includes a first interconnect access structure passing through a second portion of the dielectric pad layer to electrically connect to the CBM electrode layer. The metal-insulator-metal capacitor includes a second interconnect access structure passing through the CTM electrode layer to make physical contact with a capping layer, where the second interconnect access structure passing through the CTM electrode layer causes the second interconnect access structure to electrically connect to the CTM electrode layer above another portion of the dielectric pad layer, and where the first interconnect access structure and the second interconnect access structure electrically couple the metal-insulator-metal capacitor to the integrated circuitry.

[0094] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and layers for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A device comprising:

a first capacitor bottom metal electrode layer;

a dielectric pad layer, over the first capacitor bottom metal electrode layer, comprising one or more trench regions exposing one or more portions of the first capacitor bottom metal electrode layer; and

a second capacitor bottom metal electrode on the dielectric pad layer and the one or more portions of the first capacitor bottom metal electrode layer.

2. The device of claim **1**, further comprising:

an interconnect structure, adjacent to the second capacitor bottom metal electrode, extending through the dielectric pad layer and into the first capacitor bottom metal electrode layer.

3. The device of claim **1**, further comprising:

an insulating layer over the dielectric pad layer and the one or more portions of the first capacitor bottom metal electrode layer.

4. The device of claim **1**, further comprising:

a capping layer over the dielectric pad layer and the one or more portions of the first capacitor bottom metal electrode layer.

5. The device of claim **4**, further comprising:
a capacitor top metal electrode layer over the capping layer.

6. The device of claim **5**, further comprising:
an interconnect structure extending through the capacitor top metal electrode layer and to the capping layer.

7. A device, comprising:
a capacitor bottom metal electrode layer;
a dielectric pad layer, over the capacitor bottom metal electrode layer, comprising a region exposing a portion of the capacitor bottom metal electrode layer;
a capacitor top metal electrode layer over the dielectric pad layer and the portion of the capacitor bottom metal electrode layer; and
an interconnect structure, adjacent to the capacitor top metal electrode layer, extending through the dielectric pad layer and the capacitor bottom metal electrode layer.

8. The device of claim **7**, comprising:
an insulating layer over the dielectric pad layer and the portion of the capacitor bottom metal electrode layer,
wherein the capacitor top metal electrode layer is further over the insulating layer.

9. The device of claim **7**, wherein the capacitor top metal electrode layer resides directly on the dielectric pad layer and the portion of the capacitor bottom metal electrode layer.

10. The device of claim **7**, further comprising:
a second interconnect structure extending into the capacitor top metal electrode layer.

11. The device of claim **10**, further comprising:
a dielectric layer over the capacitor top metal electrode layer,
wherein the second interconnect structure extends through the dielectric layer and into the capacitor top metal electrode layer.

12. The device of claim **7**, further comprising:
an inter-layer dielectric layer over the dielectric pad layer and the capacitor top metal electrode layer.

13. The device of claim **12**, wherein the interconnect structure extends through the inter-layer dielectric layer.

14. A method, comprising:
forming a dielectric pad layer, over a capacitor bottom metal electrode layer, comprising a region exposing a portion of the capacitor bottom metal electrode layer;
forming a capacitor top metal electrode layer over the dielectric pad layer and the portion of the capacitor bottom metal electrode layer; and
forming an interconnect structure, adjacent to the capacitor top metal electrode layer, extending through the dielectric pad layer and the capacitor bottom metal electrode layer.

15. The method of claim **14**, comprising:
forming an insulating layer over the dielectric pad layer and the portion of the capacitor bottom metal electrode layer,
wherein the capacitor top metal electrode layer is further over the insulating layer.

16. The method of claim **14**, wherein the capacitor top metal electrode layer resides directly on the dielectric pad layer and the portion of the capacitor bottom metal electrode layer.

17. The method of claim **14**, further comprising:
forming a second interconnect structure extending into the capacitor top metal electrode layer.

18. The method of claim **17**, further comprising:
forming a dielectric layer over the capacitor top metal electrode layer,
wherein the second interconnect structure extends through the dielectric layer and into the capacitor top metal electrode layer.

19. The method of claim **14**, further comprising:
forming an inter-layer dielectric layer over the dielectric pad layer and the capacitor top metal electrode layer.

20. The method of claim **19**, wherein the interconnect structure extends through the inter-layer dielectric layer.

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